Gesture size affects numerical estimates in quantifier comprehension

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Abstract

People think and talk about numerical magnitude in terms of space, and co-speech gestures reflect this, with English speakers using expansive gestures when talking about greater quantities. Existing gestural research on the spatial conceptualisation of number has largely looked at gesture production, but we do not know whether gestures can influence the interpretation of imprecise or underspecified numerical expressions such as quantifiers. Looking at the quantifier *several* as a test case, this study investigates the influence of manual inwards-directed (i.e. hands move closer the torso) and outwards-directed (i.e. hands move away from the torso) co-speech gestures on comprehenders' conceptualisation of quantities associated with *several* through three pre-registered experiments. Our results suggest that gesture modulates the interpretation of *several* such that speakers moving their hands outwards and thereby creating space between their hands were interpreted to mean higher quantities than speakers not gesturing, and speakers moving their hands inwards and thereby restricting space between their hands. We discuss the implications of our findings for future work in numerical cognition, multimodal communication and pragmatics.

Keywords: manual co-speech gestures, quantifier interpretation, multimodal communication, cognition

1 Introduction

English speakers choose among a large repertoire of linguistic resources to communicate numerical information ranging from exact figures (e.g., "There were 3,245 people at the protest") to imprecise or vague quantifiers (e.g., "There were many people at the protest"). While exact figures map to a precise number or proportion of objects, quantifiers such as *several* depict varying ranges of numerical values that are not constant across individuals and contexts but instead vary (e.g. Bach, 2012; Coventry, Cangelosi, Newstead & Bugmann, 2010; Cummins, 2015). Language users may produce quantifiers when giving approximate quantities is less effortful than counting, or when

producing an exact number is not relevant to the speakers' communicative goals. For example, for a speaker producing (1), it might not have been relevant to specify the exact number of students who attended. Instead, in order to imply that the lecture was not a success, it is sufficient to indicate that there was a low number of students through using the imprecise quantifier *a few*.

(1) Few students attended the statistics lecture.

From a comprehenders perspective, inferring the quantity a speaker meant to convey by producing a given quantifier is known to be a complex process depending on multiple contextual factors. For example, comprehenders are found to rely on visual information such as the size of the quantified objects when interpreting quantifiers (c.f. Hörmann, 1983; Newstead & Coventry, 2000). This study investigates whether visual information in the form of manual, horizontal inwards-directed (i.e. hands move closer to the torso) or outwards-directed (i.e. hands move away from torso) co-speech gestures can serve comprehenders as further contextual cues when interpreting imprecise, quantity expressions such as quantifiers.

2 Theoretical Background

2.1 Quantifiers

Quantifiers such as *many, several, much, a lot* do not correspond to exact numerical values that are constant across individuals and contexts (Coventry et al., 2010; Cummins, 2015; Moxey & Sanford, 1993a, 1993b; Newstead & Coventry, 2000). For example, one of the few studies looking at quantifier production by van Tiel, Franke and Sauerland (2021) suggests that while some English speakers produce *several* to describe relatively small quantities – such as 80 out of a set of 432 – others produce *several* to describe more extreme values from as low as 9 to as high as 361 out 432.

While production data of quantifiers is relatively sparse, the comprehension of quantifiers is well studied and the meaning of quantifiers is found to be influenced by multiple contextual factors. For example, Newstead, Pollard and Riezebos (1987) show that the meaning of quantifiers in terms of

proportions is not constant across set sizes: participants inferred quantifiers associated with relatively low ranges of quantities such as *several* and *few* to mean higher proportions when the set size was relatively small (e.g. 60) as opposed to relatively large (e.g. 1000). Furthermore, the meaning of quantifiers has been found to be affected by the comprehenders' general expectations regarding the number/frequency of the quantified objects. For instance, Moxey and Sanford (1993b) found that quantifiers which usually denote large proportions, such as *many*, were judged to denote smaller proportions when the participants' expected proportion was small – e.g. proportion of local doctors that are female (relatively low) versus proportion of people enjoying a Christmas party (relatively high). Similarly, Moxey and Sanford (1993a) discovered that the quantifier *some* was associated with smaller quantities when, for example, describing the number of people gathering in front of a fire station as opposed to describing the number of people gathering in front of a cinema.

Quantifier meaning has additionally been found to be influenced by the extralinguistic context, such as concomitantly presented visual information (Hörmann, 1983; Newstead & Coventry, 2000). For example, when shown a picture of a bowl containing many balls, participants' interpretations of quantifiers such as *several* were found to change as a function of the number of balls shown (Newstead & Coventry, 2000). Moreover, even though the balls' size should be irrelevant in this task, the same quantifiers yielded different responses depending on whether the depicted balls were large or small: participants rated low magnitude quantifiers, such as *few*, as being appropriate for depicting relatively large numbers of balls as long as the balls were small in size. This finding is in line with Hörmann's (1983) study where low magnitude quantifiers were found to be appropriate for large numbers of small entities such as crumbs.

Overall, these findings contribute to the more general observation that language comprehension is influenced by visual context (Degen & Tanenhaus, 2016; Spivey, Tanenhaus, Eberhard & Sedivy, 2002). Given the fact that quantifier interpretation is modulated by visual context, we would expect that viewing manual co-speech gestures produced together with quantifiers should also affect

numerical estimates. Furthermore, as suggested by the research discussed above, quantifier meaning is highly imprecise which makes studying the potential influence of gestures on the interpretation of quantifiers a good starting point for studying the role of gestures in numerical comprehension in general.

2.2 Gestures

When people talk about quantities they usually also gesture. Gestures are in fact found to be abundant in numerical discourse, in political contexts and in the news (Winter, Perlman & Matlock, 2013). Findings of Woodin, Winter, Perlman, Littlemore and Matlock's (2020) large-scale analysis of more than 500 English speakers suggest that adults in the TV news move their hands outwards (i.e. away from their torso) when talking about large quantities and inwards (i.e.towards their torso) when talking about small quantities providing evidence for an association between numerical quantity and spatial extent. In this study, we focused on gestures such as these that communicate size, for example, hands moving far apart to indicate a large size, or thumb and index finger 'pinching' to indicate a small size (Hassemer & Winter, 2016, 2018; Sowa, 2006).

In using a size-related gesture to investigate people's inferred numerical estimates when interpreting *several*, this study also investigates the interaction between visually presented spatial extent and quantity more generally. It is known that the size of visually presented information, such as the size of visually presented numerals (e.g. Henik & Tzelgov, 1982) or the area covered by dots in dot patterns (e.g. Hurewitz, Gelman & Schnitzer, 2006), interacts with people's internal magnitude representations. This has also been shown with respect to size-related hand actions, such as for the contrast between precision grips (index finger and thumb together) used for grasping small and light objects and power grips (a hand shape as if holding a pipe) used for grasping large and heavy objects (Napier, 1956).

So far gestures have been found to play a role in the production of numerical information as well as

children's learning of mathematical concepts (Alibali et al., 2013; Goldin-Meadow, Cook & Mitchell, 2009; Goldin-Meadow, 2014; Novack & Goldin-Meadow, 2015; Goldin-Meadow, Nusbaum, Kelly & Wagner, 2001). Research on the role gesture might play in the comprehension of numerical information in adult-to-adult communication has so far been relatively sparse. A study by Nicol and Patson (2022) attested the influence of gestures on comprehenders' interpretation of plurals. Using a picture matching paradigm, Nicol and Patson (2022) found that participants preferred pictures containing a relatively small set of objects when the plural noun was accompanied by an inward gesture, and likewise, preferred pictures containing a relatively large set of objects when the plural noun was accompanied by an outward gesture. This effect is in line with what we would expect to find in connection with quantifiers: people should infer larger numbers when seeing an outwards-directed gesture and smaller numbers when seeing an inwards-directed gesture when interpreting quantifiers.

3 The present study

We conducted a total of three pre-registered experiments with the aim of investigating whether horizontal outwards- and inwards-directed gestures – modelled after gestures attested in footage of English-speaking TV news anchors talking about quantity/ numerical information (Woodin et al., 2020) – affect the interpretation of *several*. We specifically focused on the quantifier *several* since it was found to be associated with an extremely broad range of quantities: van Tiel et al. (2021) found that *several*, *some* and *many* were produced for broader ranges than *most*. We were expecting that a quantifier with a broader rather than a narrower range of interpretations might be more susceptible to the influence of gesture, which makes *several* a good starting point for probing whether quantifier interpretation is malleable to gesture.

In order to investigate the influence of gesture on the interpretation of *several*, we manipulated the direction of gestures accompanying verbal instances of *several* which correlates with their spatial magnitude, as discussed in more detail later. We predicted the following relative ranking of inferred

quantities for *several* in the three conditions: inward < no < outward. In order to directly assess whether participants rely on the association between space and numerical magnitude when interpreting gestures in conjunction with *several*, we further tested whether the gestures' size (distance between the gesturer's hands) affected participants' responses. This series of experiments proceeded as follows. With experiment I, a one-trial between-subjects experiment, we investigated the effect of gesture on the interpretation of *several* as a between subjects factor, mirroring a communicative context in which comprehenders see a speaker producing a gesture on TV without knowing this speaker's usual gestural behaviour. Experiments II and III investigated the effect of gesture as a within-subjects factor. This study received research ethics committee approval.

4 Experiment I

4.1 Linguistic Materials

Experiment I investigated the influence of co-speech gestures on the interpretation of utterances involving the quantifier *several* embedded in different discourse contexts. We created 20 stimuli consisting of a context sentence followed by a quantified statement (i.e. the manipulated sentence in which a given quantifier co-occurred with a given gesture). For example (2) illustrates two stimuli, each with a context sentence (*400 people were at the protest/400 people participated in the trial*) followed by a quantification sentence (*Several of them got arrested/Several of them experienced side effects*).

(2) a. 400 people were at the protest. <u>Several of</u> them got arrested.

b. 400 people participated in the trial. <u>Several of</u> them experienced side effects.

The overall set size used for all sentences was 400 in order to be able to compare our results with van Tiel et al.'s (2021) results, who used 432 as overall set size (i.e. the number of presented balls). We chose the round number 400 instead of 432 to ease the participant's reasoning process when performing the task: since round numbers are likely interpreted as approximations (Krifka, 2007),

we believe that using an even set size aligns more closely with the participants' task to interpret what they think *several* means and avoids the impression that there is a single correct response. We decided to keep the set size constant across sentences due to Newstead et al.'s (1987) findings suggesting that the interpretation of quantifiers in terms of proportions is not constant across set sizes.

Numerical language is relevant to a whole range of different discourses (e.g., finance, politics, health care). In alignment with concerns about the generalizability of research in psychology (Yarkoni, 2020), we assessed the influence of gesture on the interpretation of *several* for a large range of different discourse contexts, in order to avoid limiting our conclusions to any one particular context. We are thereby demonstrating that the influence of gesture on the interpretation of *several* is general, happening regardless of the discourse context in which they feature. Thus, each item used a different context, from different topical domains, including politics (2a), and science (2b), see Appendix I for all experimental stimuli.

4.2 Video Materials

We video recorded two British English speakers in frontal-view, both lecturers at the University of Birmingham, while producing the earlier introduced linguistic materials with either no gesture, or inwards-/outwards-directed gestures. Figure 1 illustrates the end-points for both gesture conditions after the inwards- (top) or outwards-directed (bottom) movement. The gestures were modelled after the inward and outward gestures which were found to correlate with the phrase *large/huge number* and *tiny/small number* in Woodin et al. (2020). The speakers were instructed to execute the primary inwards or outwards movement together with the underlined phrase, such that the stroke phase (the primary movement phase of a gesture) was aligned with the relevant quantification sentence. Following this, we instructed speakers to perform a post-stroke hold so that a return to resting position did not create interference (e.g., hands moving inwards). The speakers were

allowed to vary their gestures within reason but were instructed to have their hands elevated to torso height. We return to this variation in gesture production in our statistical analyses later. Each context-quantification sentence pair was recorded in full separated by a small pause.

Figure 1: Example for an inward (left column) and outward (right column) gesture produced by Speaker1 (top-row) and Speaker 2 (bottom-row) in resting position after the stroke phase. Arrows indicate the direction of the gesture's stroke phase (outwards- versus inwards-directed).

4.3 Procedure

The experiment started with four practice trials during which we presented participants with four sentences in text form that were formulated like our experimental items and contained the number 400 and the quantifier *several* (e.g. *400 books were gifted to charity. Several of them were novels*). Like for the experimental trials, participants were asked to estimate what number *several* indicates. We introduced these practice trials in order to familiarise participants with the task and ensure that

the quantifier was not misheard in the audio presentation: in a pilot study[1] participants mistook several for seven.

The three-level factor gesture was tested as a between-subjects factor, such that each participant was randomly assigned to one experimental item in only one condition. This, as opposed to multiple trial experiments or a within-subjects manipulation, was done to maintain ecological validity. Being confronted with a speaker who produces a quantifier in conjunction with gestures in a conversation or during a speech most plausibly happens as an individual occurrence rather than a series of occurrences. Our experimental design reflects that. After having seen the video of a speaker producing the context and quantification sentence (quantifier + gesture), participants were asked to make a numerical estimation by typing a number into a text field. For example, having heard (2a) or (2b) participants were asked "Guess how many of the 400 people got arrested?" or "Guess how many asylum seekers were young children?" respectively. We specifically asked participants to type in a numerical value.

After the experimental trial, participants were asked two follow-up questions: "What do you think has influenced your judgement when answering the questions about the videos?" with a text-response and "To what extent did you notice the speakers' hand movements in the videos?" with a 5-point Likert scale response ("Definitely not" – "Definitely yes").

We included two attention checks in the experiment which consisted of two questions based on the method by Kennedy, Clifford, Burleigh, Jewell and Waggoner (2018): participants were asked to specify their birth year at the beginning and at the end of the experiment. Participants whose answers did not correspond to each other were excluded.

[1]See OSF repository Mhttps://osf.io/zwsf8/?view_only=cca14ab2e19c4321bb161b98429fc240.

4.4 Participants

Participants (N=600) were recruited via Prolific and our sample was restricted to native speakers of British English and those with an approval rate above 90%. The sample size was based on findings of a between-subjects pilot study which initially suggested that the influence of gesture on quantifier interpretation may be small (difference of 5% between outward and inward condition). With a total of 600 participants we tested approximately 200 participants per condition and 15 data points per experimental item. We asked for the participants' age and self-identified gender to characterise our sample. Participants were paid £1 which amounted to £17.5/hour (the average duration of the experiment was 3:42 minutes).

After data collection, we excluded participants who did not pass our attention check, those that had reported to have technical difficulties (e.g. video was not shown/sound not audible), those that did not specify to be a native speaker of English, and those that reported values outside of the range constrained by the context set which has an upper limit of 400. This led to the exclusion of 19 data points (3.2%). After data exclusion, 581 participants remained as part of our sample. The age of the remaining participants ranged from 19 to 78 years, with a mean of 42 years (median = 40). 7 participants stated their preferred pronoun as they/them, 281 participants as she/her, 278 chose he/him, 10 chose the option "other", and 5 participants chose not to disclose this information.

4.5 Analysis

All analyses were pre-registered (`Mhttps://osf.io/zwsf8/?view_only=cca14ab2e19c4321bb161b98429fc240`) and conducted with R (R Core Team, 2020) and the tidyverse package (Wickham et al., 2019). The main response variable in our study was the proportion of the total set size. We modelled this using a Bayesian beta binomial regression model implemented via the brms package (Bürkner, 2017). To guide our choice of a generalized linear model, we first modelled the production data from van Tiel et al. (2021) with various candidate models. Via posterior predictive simulations, we established that the beta binomial distribution most plausibly could have generated data of this kind (see OSF re-

pository Mhttps://osf.io/zwsf8/?view_only=cca14ab2e19c4321bb161b98429fc240). We parametrized the beta binomial distribution in terms of the parameters μ (mean) and ϕ (precision), in analogy to mean and variance for the normal distribution. This allowed us to model whether the gesture condition leads to differences in mean estimates, but also whether estimates are more or less variable across conditions.

In total we ran three models to investigate different aspects of our data. With Model 1, we investigated the influence of gesture on the interpretation of *several*, by comparing the gesture conditions (inward gesture/no gesture/outward gesture) with each other. In addition to the influence of gesture, for Model 2, we included speakers as a factor, since one cannot assume that our two speakers spoke and gestured the same. With Model 3 we directly assessed whether participants rely on the association between space and magnitude when interpreting gestures in conjunction with *several*. For this purpose, we measured the horizontal distance in pixels between the speakers' hands when the gesture reached resting position by using the MATLAB Image Viewer App (MATLAB, 2022), excluding the no gesture condition. This analysis allowed us to make use of the natural variation between the way speakers produce gestures to test the idea that numerical estimates is metrically related to the size of the gesture, i.e., relatively larger outward gestures with more extended arms could lead to higher estimates than less expansive gestures in the same condition.

Model 1 only included the experimental factor gesture (inward/no/outward) to predict the proportion of the total set size. The factor gesture was treatment-coded with no-gesture as reference level. We modelled ϕ as a function of the gesture condition. This allowed for the possibility that some of the gesture conditions had more or less variance in responses, i.e., wider or narrower distributions of numerical estimates. Model 1 included random intercepts for item (20 levels), and since in this design, a unique speaker-item combination is also a repeated factor (e.g., a group of participants saw the female speaker produce sentence (2a), for example), we included speaker-item combination (40 levels) as an additional random intercept. There were no by-participant random intercepts or ran-

dom slopes in this model because each participant only gave one response (between-subjects design).

Model 2 included the factor speaker (Speaker1/Speaker2) and the interaction between gesture and speaker as predictors. To facilitate the interpretation of main effects in the presence of interactions, we sum-coded the predictors speaker (Speaker1=1/Speaker2=-1) and gesture, see Table 11 in Appendix II for the gesture contrast coding. Model A2 involved random intercepts and slopes for items following the same reasoning as for Model 1 and ϕ was specified to vary for gesture and speaker.

Model 3 included the continuous predictor pixel distance representing the gesture's size. The model included random intercepts and slopes for items accounting for the possibility that the gesture effect may vary between items and ϕ was specified to vary by pixel distance.

We used the default priors of brms except for the priors of the intercept, main condition effect as well as the precision parameters. We used regularising priors, which allowed a reasonably wide range of parameter values and at the same time penalise very extreme values. The prior for the intercept was a normal distributed with mean 0 and standard deviation 1. Furthermore, to build mild scepticism into our models (Lemoine, 2019; McElreath, 2020), we set a weakly informative prior on the condition effect: a normal distribution centred at zero with a standard deviation of 1. Assuming a logit intercept of zero, this prior embodies the assumption that 95% of all differences range from $p = 0.12$ to $p = 0.88$.

Samples were drawn from the posterior distributions of the model parameters using the NUTS sampler (Hoffman & Gelman, 2013). We ran four sampling chains, each collecting 4000 iterations whereby the first 1000 iterations were disregarded as part of the warm-up phase leading to 12,000 posterior samples available for analysis. Chains mixed well (all $\hat{R} = 1.0$). As inference criterion, we focused on reporting the posterior estimates and 95% Bayesian credible intervals for fixed effects and interaction terms. In addition, we computed the posterior probability of the effect being the

same sign. For all models, the posterior distributions over population estimates and estimates of random effects can be found in Appendix II.

4.6 Results: Model 1

The estimated quantities ranged from 2 to 400, with a mean of 81.2 and a standard deviation of 92.7. Figure 2 shows the probability density distribution over quantities assigned by participants. Similar to van Tiel et al.'s (2021) findings, the distribution over quantities for *several* was a broad, right-tailed distribution, suggesting that the meaning of *several* is most frequently interpreted as depicting quantities below 100. Figure 2 also suggests an overall preference for round numbers for higher magnitudes, such as 100, 150, 200 etc. (Dehaene & Mehler, 1992; Jansen & Pollmann, 2001; Woodin, Winter, Littlemore, Perlman & Grieve, 2023).

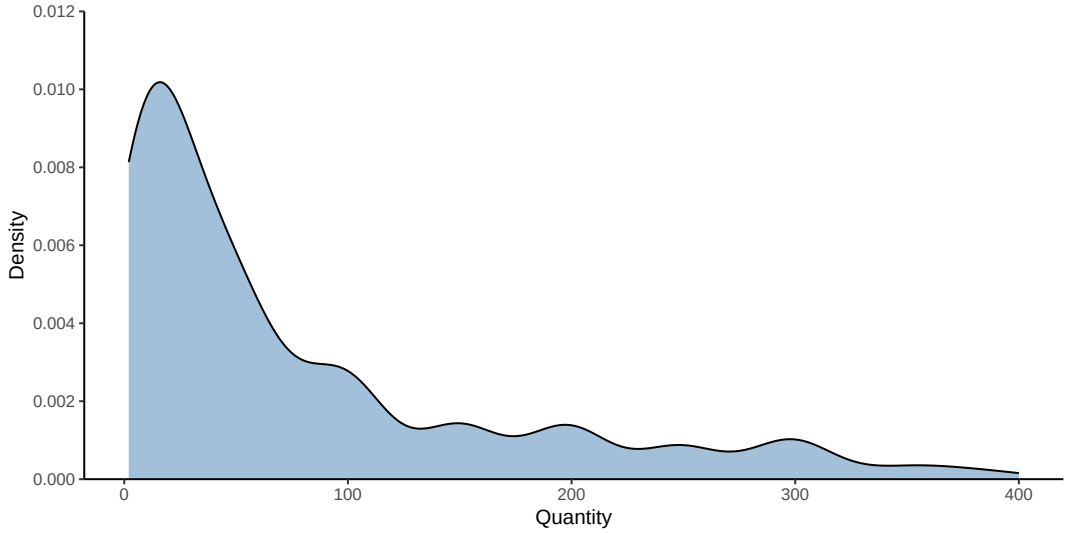


Figure 2: Probability density plot for *several*.

Considering just the trials in which participants were exposed to a speaker producing either one of the two gestures, participants (N=369 out of 581) generally reported to have definitely noticed the gesture, see Table 1.

Response	Absolute counts
Definitely not	2
Not really	19
May or may not	3
To some extent	52
Definitely yes	293

Table 1: Response counts for the 5-point Likert scale question "To what extent did you notice the speakers' hand movements in the videos?" restricted to participants who saw either one of the two gestures. Responses ranged from "Definitely not" – "Definitely yes".

Most importantly, quantities assigned in the outward gesture condition (mean=97.6 , median=50.0) were on average 14.0 higher than the quantities assigned in the no gesture condition (mean=83.6, median=30) and 35.4 higher than in the inwards-directed gesture condition (mean=62.2, median=30). Quantities varied more in the outward (sd=105.0) than in the no gesture condition (sd=91.7) or in the inward gesture condition (sd=62.2), see Figure 9 in Appendix II.

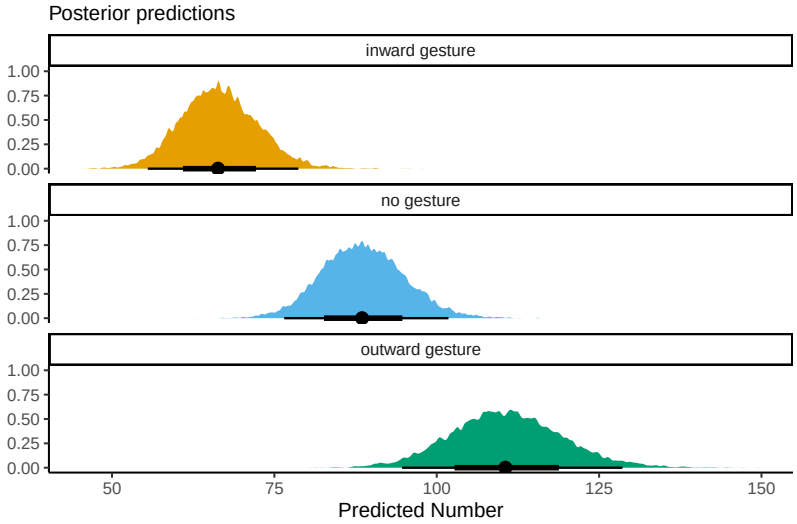


Figure 3: Expected values drawn from the posterior predictive distribution for each of the gesture conditions. The dot represents the mean and the lines represent the 0.80 and 0.95 highest-density intervals (HDI).

To quantify uncertainty in these results, our Bayesian mixed beta binomial regression (Model 1) indicated a strong effect of gesture such that participants who saw a speaker producing an inward gesture assigned lower quantities than participants who saw a speaker not gesturing (logit coefficient: b=-0.36, [-0.61, -0.11]). Participants who saw a speaker producing an outward gesture assigned the highest quantities (logit coefficient: b=0.30, [0.05, 0.55]), see Figure 3. In a further pre-registered analysis for which we excluded the no gesture condition, we also found a reliable difference between inward and outward gesture conditions, such that participants who saw a speaker producing an inward gesture assigned lower quantities than participants who saw a speaker producing an outward gesture (logit coefficient: b=0.63, [0.37, 0.90], reference level=inward), see Appendix II for the full model.

The precision coefficient ϕ for gesture suggests that the distribution over quantities in the outward gesture condition was less precise than the one for the inward gesture condition ($\hat{\beta}$ =-0.33, [-0.60, -0.06]). This is illustrated further in Figure 3 (right): whereas the expected posterior quantities in the inward gesture condition were on average lower than the quantities in the no and outward gesture conditions, the distribution over expected quantities in the outward gesture condition had a wider spread than the distributions over expected quantities in the no and inward gesture conditions. Overall, the results suggest that the interpretation of *several* is influenced by whether or not a speaker manually gestures and what kind of gesture they produce: comprehenders seeing inwards-directed gestures infer lower quantities than those who see outwards-directed gestures.

4.7 Results: Model 2

It is also possible that responses varied as a function of our two model speakers. This difference between speakers is in fact already visible in our example stimuli shown in Figure 1 above, such that one speaker (Speaker1, the male model) produced larger outward gestures than the other speaker (Speaker 2, the female model). More concretely, when measuring the horizontal distance in pixels between the speakers' hands when the speakers' gesture reached resting position (excluding the no

gesture condition), the pixel distance for Speaker1 ranged from 31 to 1472 pixels and for Speaker2 from 156 to 687 pixels. The difference was most apparent for the outwards-directed gesture, see Table 2.

Speaker	Gesture	mean	sd	median	min	max
Speaker1	inward	66.1	28.9	60.0	31	130
Speaker1	outward	1273.7	137.3	1294	948	1472
Speaker2	inward	207.4	31.9	214	156	265
Speaker2	outward	580.3	56.2	561	476	687

Table 2: Pixel distance between the speakers' hands in the resting position of the inward versus outward gesture (M=505, SD=453, range=[31, 1472])

The natural variation between these two speakers allowed us to assess whether gesture size matters with Model 2. If our results indicate that being exposed to Speaker1 led to higher/lower quantities in the outward/inward conditions than being exposed to Speaker2, we would have some evidence suggesting that the size of the gesture is positively correlated with the quantities participants assigned.

Results of Model 2 indicate a strong gesture effect such that, averaged over speakers, participants assigned highest quantities in the outward gesture condition, followed by the no and then inward gesture conditions, see Appendix II for the full model output. The model similarly revealed that, averaged over the gesture conditions, Speaker1 led to a decrease in quantities ($\hat{\beta}$ =-0.05, [-0.16, 0.05], in log-odds). The posterior probability of this effect being above zero was only 0.85 and cannot be considered reliable.

The model output is illustrated further in Figure 4: whereas the expected posterior quantities in the inward gesture condition were on average lower than the quantities in the no and outward gesture conditions, the difference is larger for Speaker1. These results suggest that – ignoring the no gesture condition for now – there is a correlation between gesture size and the assigned quantities, such

that the bigger the size of the gesture the higher the quantities.

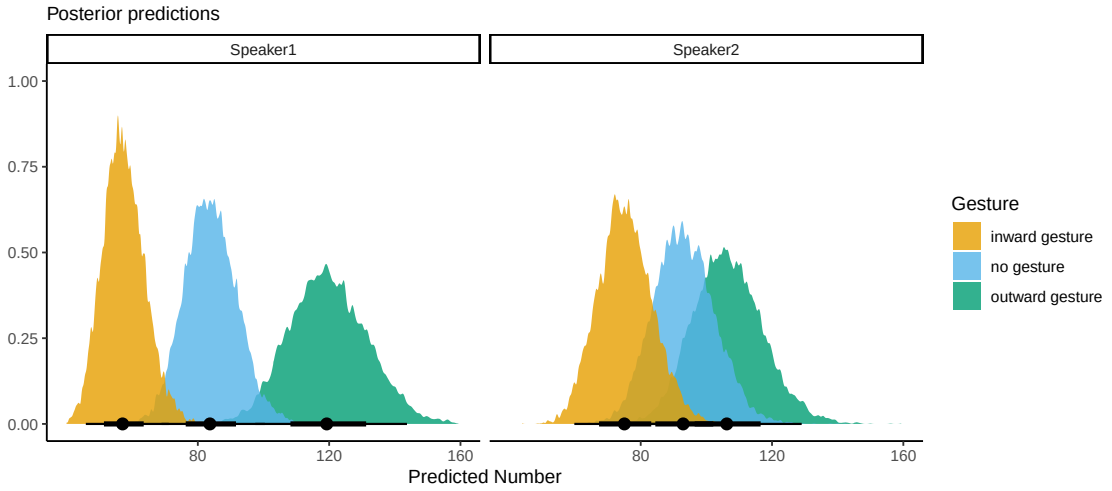


Figure 4: Expected values drawn from the posterior predictive distribution for each of the gesture conditions. The dot represents the mean and the lines represent the 0.80 and 0.95 highest-density intervals (HDI).

4.8 Results: Model 3

To evaluate the differences between speakers in terms of gesture size more concretely and to account for gesture size variation within speakers we analysed our data with Model 3, for which we included pixel distance (standardised) as a continuous predictor to predict the quantity that participants associated with *several* (excluding the no gesture condition). Model 3 indicated that an increase in one standard deviation (i.e. 453 pixels) of distance led to an increase in log-odds ($\hat{\beta}$ =0.33, [0.20, 0.47]) – translating to an increase of 24 in quantities. This means that an increase of 100 pixels led to an approximate increase of 5 in quantities. The results also suggest a decrease in precision ($\hat{\beta}$ =-0.34, [-0.49, -0.19]). See Figure 5 for an illustration of the effect of pixel distance on Quantities.

Coefficient	posterior mean	Standard Error	l-95% CrI	u-95% CrI
μ intercept	-1.29	0.07	-1.44	-1.14
μ distance (px)	0.33	0.07	0.20	0.47
ϕ intercept	1.09	0.08	0.94	1.23
ϕ distance (px)	-0.34	0.08	-0.49	-0.19

Table 3: Population-level estimates of the Bayesian beta binomial regression model on the log-odds scale with 95% credible intervals. The continuous predictor distance (px) was standardised.

These results suggest that the size of the produced gestures affected participants' interpretation of *several*: the further apart the speakers' hands were during the resting position of the gesture, the larger the inferred quantities for *several*. Similarly, the findings for ϕ suggest that the larger the gesture, the higher the variability between participants' interpretations.

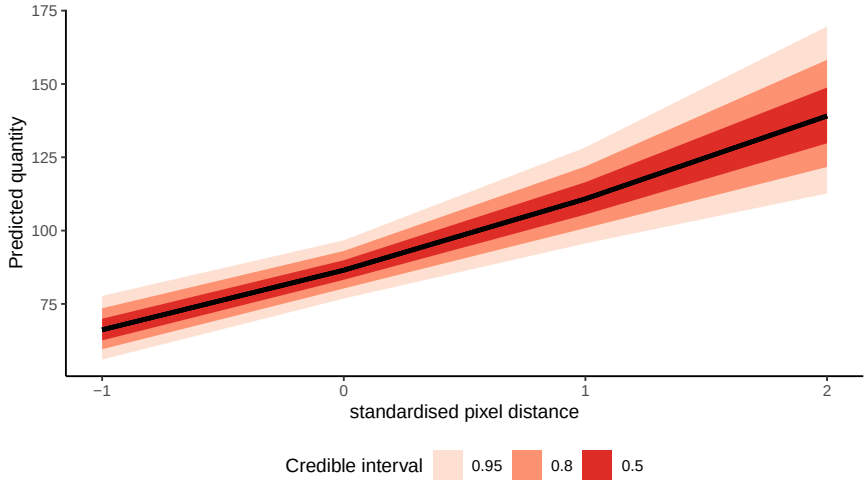


Figure 5: Average marginal effect of distance in pixels between the speakers' hands in post-stroke holding position. 0 on the x-axis corresponds to a pixel distance of 505 (SD=453, range=[31, 1472]).

4.9 Results: Differences between discourse contexts

The assigned quantities varied greatly between discourse contexts. Recall that we created 20 different discourse contexts, see (3) for examples or Appendix I for a complete list.

(3) a. 400 voters were interviewed. Several of them reported to have voted for Labour.

b. 400 people were at the protest. Several of them got arrested.

c. 400 new research projects received funding. Several of them are in Humanities.

d. 400 climate events occurred last year. Several of them were catastrophic.

e. 400 elephants were born this month. Several of them were documented in India.

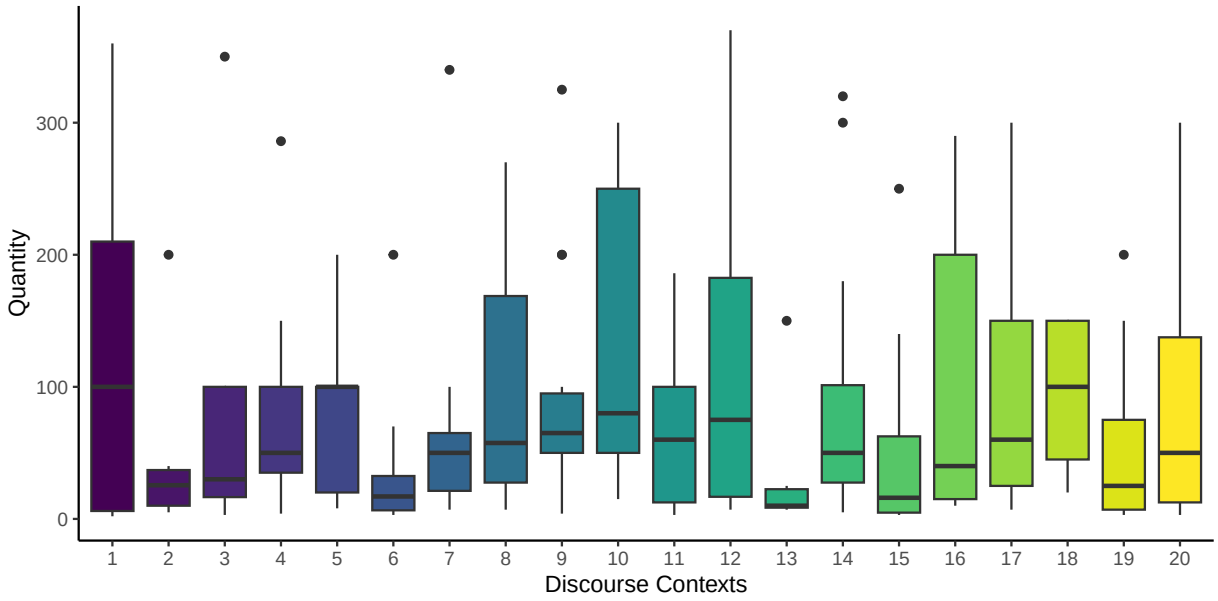


Figure 6: Difference between discourse contexts of experiment I.

In order to highlight differences between discourse contexts we focused on the no gesture condition trials of experiment I, assuming that in those trials participants mainly relied on the linguistic material when interpreting *several*. Figure 6 illustrates the median assigned quantities and corresponding upper and lower quantiles for each discourse context. For example, in context 1, i.e. (3a), participants interpreted *several* to mean most often 100, however, the distribution over quantities was very broad, such that 50% of the data lied between approximately 10 and 210, suggesting that this discourse context was interpreted quite differently between participants. In contrast, in context (2), i.e. (3b), participants assigned mostly low quantities to *several* with a median of 25.5 whereby participants' responses varied much less.

We can only speculate which factors drove the variation between discourse contexts. Taking contexts (1/3a) and (2/3b) as examples, for context (1/3a), participants may have relied on their world knowledge about how many people in general vote for Labour which then may depend on factors such as the participants' education, place of living, political preferences etc. For context (1/3b), participants may have relied on a more commonly shared impression that protests in the UK generally seem not to involve violence or other factors that could lead to arrests by the police. While these are merely speculations, our findings replicate previous research which postulated that comprehenders' prior expectations affect their interpretation of quantifiers (Moxey & Sanford, 1993a, 1993b).

5 Experiments II and III

5.1 Design, Materials, Procedure

We conducted two more pre-registered experiments investigating gesture as a within-subjects factor for three reasons: to replicate the findings of experiment I, to investigate individual differences which we cannot do for our single-trial data of experiment I, and to test communicative contexts in which a comprehender sees both the inward and outward gestures. Recall that in using a between-subjects design with a single trial, experiment I mirrors a communicative context in which comprehenders see a speaker gesturing once without knowing their usual gestural behaviour, e.g., how much they gesture, or how large the person's gestures are. In experiment II, participants saw both gestures produced by the same speaker mirroring a communicative context such as a speech in which comprehenders have more time to get to know the speaker's gesture behaviour. In experiment III, participants saw both gestures produced by a different speaker mirroring a communicative context such as a debate in which different speakers use similar manual gestures.

Comparing experiment II and III enables us to see whether the contrast between outward and in-

ward gestures relied on/was enhanced by the gestures being carried out by the same speaker versus different speakers. For both experiments, we analysed the participants' numerical estimates as a function of gesture condition, again with Bayesian beta-binomial regression models. Due to their high similarity, we discuss experiments II and III together.

As before, we investigated the influence of the three-level factor gesture (inward/no/outward) on the interpretation of utterances involving the quantifier *several* embedded in different discourse contexts. This time, gesture was tested as a within-subjects factor such that each participant saw three videos one in each of the three gesture conditions. In experiment II, participants were exposed to always the same speaker. In experiment III participants were exposed to both speakers (one speaker was seen twice and the other one once). For both experiments, participants were randomly assigned to lists to ensure that they would not see the same stimuli more than once, and for experiment III to ensure that participants would see both speakers. Stimuli and attention checks were the same as in experiment I. The only deviation from experiment I regarding the procedure is that participants saw three videos in total.

5.2 Participants

Participants in experiments II and III (N=204/199[2] respectively) were recruited via Prolific and our sample was restricted to native speakers of British English and who had an approval rate above 90%. The sample size was based on findings of a within-subjects pilot study[3] which indicated that the influence of gesture on quantifier interpretation is relatively large in comparison to the between-subjects pilot. We asked for the participants' age and self-identified gender to characterise our sample. Participants were paid £1.16 which amounted to £17.5/hour (the average duration of the experiments was 4:00 minutes).

[2]We pre-registered 200 participants per experiment and the deviation in sample size happened during the recruitment process.

[3]See OSF repository Mhttps://osf.io/zwsf8/?view_only=cca14ab2e19c4321bb161b98429fc240.

After data collection, we excluded participants who did not pass our attention check, those who reported having had technical difficulties (e.g. video was not shown/ sound not audible), those that did not specify to be a native speaker of English, and those who reported values outside of the range constrained by the context set which had an upper limit of 400. After data exclusion 200 (experiment II) and 193 (experiment III) participants remained.

For experiment II, the participants' age ranged from 18 to 74 years, with a mean of 38 years (median = 38). 3 participants stated their preferred pronoun as they/them, 105 participants as she/her, 86 chose he/him, 4 chose the option "other", and 2 participants chose not to disclose this information. The participants' age in experiment III ranged from 19 to 81 years, with a mean of 40 years (median = 37). 1 participant stated their preferred pronoun as they/them, 95 participants as she/her, 88 chose he/him, 5 chose the option "other", and 4 participants chose not to disclose this information.

5.3 Analysis

We conducted the same analyses as for experiment I, but we only discuss Model 1 and Model 3 in detail, see Appendix III for full model outputs including the one for Model 2. However, because we are now dealing with a within-subjects repeated measures design, the question of random slopes for participants poses itself. We therefore added by-participant varying slopes for the condition effect to the model (Model 1 and 3, pre-registered). This explicitly models the fact that the gesture effect may be stronger or weaker for particular participants. For all models, we had to exclude random slopes for participants due to convergence issues with $\hat{R}$ much in excess of 1.0 and the Bulk Effective Samples Size (ESS) being too low. Presumably, there were not enough data points within each participant to estimate slopes reliably.

5.4 Results: Model 1

The estimated quantities for experiment II (within-subjects, same speaker)/experiment III (within-subjects, different speaker) ranged similarly as for experiment I from 0 to 390/0 to 400 respectively,

with means of 87.9/75.4 and standard deviations of 92.6/85.9. Overall, participants noticed the speakers' gestures, see Table 4.

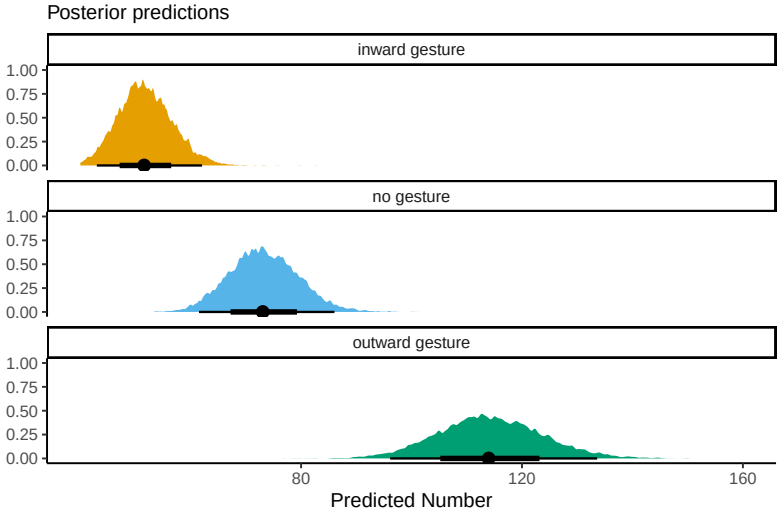
Response	Ex II	Ex III
Definitely not	2	1
Not really	8	4
May or may not	2	1
To some extent	32	33
Definitely yes	156	154

Table 4: Response counts for the question "To what extent did you notice the speakers' hand movements in the videos?" (experiments II and III).

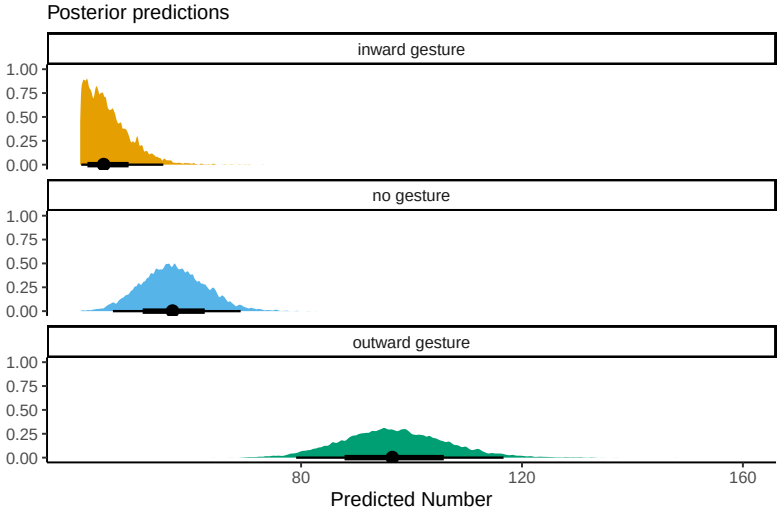
We replicated the results of experiment I and found that for both experiments II (within-subjects, same speaker) and III (within-subjects, different speaker), participants assigned lowest quantities when seeing a speaker producing an inward gesture (logit coefficient: $\hat{\beta}$ =-0.41, [-0.61 , -0.22]/$\hat{\beta}$ =-0.37, [-0.63, -0.10]), and highest quantities when seeing the same or a different speaker producing an outward gesture (logit coefficient: $\hat{\beta}$ =0.58, [0.37, 0.79]/$\hat{\beta}$ =0.66, [0.43, 0.88]). In a further pre-registered analysis[4] for which we excluded the no gesture condition, a reliable difference between inward and outward gesture condition was replicated for both experiments II (within-subjects, same speaker) and III (within-subjects, different speaker): when participants saw a speaker producing an inward gesture they assigned lower quantities than when seeing a speaker producing an outward gesture (logit coefficient: $\hat{\beta}$ =0.79, [0.54, 1.03]/$\hat{\beta}$ =0.80, [0.54, 1.06], reference level=inward). Back to Model 1, the precision coefficient ϕ for gesture suggests that the distributions over quantities in the inward gesture condition were again more precise than in the other conditions ($\hat{\beta}$ =0.61, [0.18, 1.07]/$\hat{\beta}$ =0.47, [-0.03, 0.97]). See Appendix III for more details on the model output.

The posterior predictions are captured by Figure 7. Overall we found differences between the three

[4]Due to further convergence issues we also had to exclude varying by-participant intercepts.



(a) Experiment II



(b) Experiment III

Figure 7: Expected values drawn from the posterior predictive distribution for each of the gesture conditions. The dot represents the mean and the lines represent the 0.80 and 0.95 highest-density intervals (HDI).

experiments which suggest that the influence of gesture depends on the communicative context. While responses of experiment III were overall lower than those of experiment II – perhaps due to the specific pool of participants preferring to interpret *several* to mean lower quantities than participants of experiment II – responses of experiment III also varied more within each gesture condition than those of experiment II. This variation suggests that comprehenders seeing the same speaker repeatedly agree more on the way they take gesture into account when interpreting *several* perhaps because the contrast between gestures gets more apparent when being performed by the same speaker. Moreover, in comparison with experiment I, the differences between the conditions – especially between the inward and outward conditions – were more pronounced for experiments II and III. These differences may indicate that seeing a specific type of gesture, here horizontal co-speech gestures, repeatedly may increase the extent to which comprehenders rely on these gestural cues when interpreting numerical information conveyed by the quantifier *several*.

5.5 Results: Model 3

We replicated the results of experiment I for both experiments. The model suggested that an increase in one standard deviation of distance led to an increase in log-odds ($\hat{\beta}$ =0.49, [0.37, 0.61])/($\hat{\beta}$ =0.50, [0.36, 0.65]), as well as a decrease in precision ($\hat{\beta}$ =-0.44, [-0.65, -0.25])/($\hat{\beta}$ =-0.55, [-0.76, -0.34]). These results suggest that gesture size affected participants' interpretation of *several*: the further apart the speakers' hands were during the resting position of the gesture, the larger the inferred quantities for *several*. Similarly, the findings for ϕ indicate that the bigger the gesture the higher the variability between participants' interpretation.

5.6 Results: individual differences

Overall, participants' responses varied along two previously discussed dimensions, i.e. discourse contexts and communicative contexts. Here, we will introduce individual preferences as a third dimension of variance. Considering individual differences is motivated by findings in numerical cognition which suggest high variability between cognitive processes, for example, it has been es-

tablished that not everyone shows the spatial-numerical association of response codes (SNARC) effect (Cipora et al., 2019). Given that our participants were exposed to each gesture condition only once, we cannot make strong claims about the effect of gesture on individual participants across contexts. Thus, additional research is necessary to investigate individual variances in the impact of gestures by aggregating individual responses across contexts in each of the gesture conditions. Our current assertions are tentative in nature.

We focus on data of experiment II and show how individual participants were affected by gestures executed by the same speaker. We excluded the no gesture condition and participants for which the outward and inward gesture videos did not immediately follow each other (i.e. cases in which the no gesture condition intervened). Figure 8 captures the assigned quantities for each participant seeing the same speaker producing an inward versus an outward gesture. Each line represents one participant grouped depending on whether gesture affected their interpretation of *several* negatively (inward>outward), neutrally (inward=outward) or positively (inward<outward). Participants were grouped further depending on whether they saw first the inward gesture (in, top row) or first the outward gesture (out, bottom row).

Figure 8: Individual difference between participants of experiment II. Each of the lines represent one participant. Participants were assigned into groups highlighting whether they assigned lower (inward>outward), the same (inward=outward) or higher (inward<outward) quantities in the outward as opposed to the inward gesture condition; and whether participants saw the inward (top row) or outward gesture first (bottom row) first.

Figure 8 highlights the participants' individual preferences when interpreting *several* and how gesture affected their responses. Comparing data points within a given condition, the figure illustrates individual preferences in what *several* means in terms of quantities. For instance, participants who saw an inward gesture first (top row), assigned quantities ranging from 0 to 300 in the inward condition. Furthermore, we can see – in alignment with our overall results – that the majority of participants (73.1%) show a positive trend such that they assigned higher quantities in the outward than the inward condition (inward<outward) regardless of whether they saw the inward or outward gesture first. For 13.7% of the participants gesture did not have an effect (inward=outward), and

13% of the participants showed the opposite trend (inward>outward). Even within these 'trend' groups, the strength with which the factor gesture affected participants varied, as can be seen by the variation in slopes.

Whether or not and what kind of effect gesture had on participants' estimates might have depended to some extent on whether or not participants noticed the gestures. Recall that we asked participants whether they noticed the gesture on a 5-point Likert scale ranging from "Definitely not" to "Definitely yes". While 81.1% of the participants who assigned higher quantities in the outward than inward condition (inward<outward) reported to have definitely noticed the gestures, only 73.6% and 55.5% of the participants who were either not affected by gesture (inward=outward) or displayed the opposite effect (inward>outward) reported to have definitely noticed the gestures respectively. Figure 8 similarly reveals that participants who saw the outward (bottom row) as opposed to the inward gesture first (top row), more frequently showed a negative (inward>outward) or no effect (inward=outward). This suggests that the order in which participants saw the inward and outward gestures in experiments II and III introduced subtle differences.

6 Discussion

6.1 Conceptualisation of quantity in terms of space

The results of all three experiments indicate that manual co-speech gestures of magnitude – which were found to be produced in abundance by English speakers communicating numerical information (Woodin et al., 2020) – are not simply a by-product of speech but matter for comprehenders when interpreting (imprecise) numerical information. More concretely, our findings suggest that when interpreting *several* in conjunction with gestures, comprehenders associate manual gestures that create space on the horizontal axis with relatively high quantities, and those that restrict space with relatively low quantities. The quantities comprehenders associate with *several* did not only depend on the direction of the gesture but also on the size of the gesture produced, i.e. how much

space was created or constricted by the speakers' hands.

Thus, our results provide evidence for a connection between spatial-numerical associations and gesture that is in line with studies that have investigated how the size of visually presented information, such as the size of visually presented numerals (e.g. Henik & Tzelgov, 1982) or the area covered by dots in dot patterns (e.g. Hurewitz et al., 2006), interacts with people's internal magnitude representations. Moreover, our results correspond with recent research exploring the impact of spatial gestures on numerical communication. Similar to the findings of Nicol and Patson (2022), which suggest that spatial gestures influence the interpretation of plurals, our study indicates that spatial gestures also influence the interpretation of imprecise quantifiers. This observed gesture effect on quantifier interpretation resonates with recent discoveries regarding the impact of non-manual gestures, such as facial movements, on quantifier interpretation: (Nölle et al., 2022) found that comprehenders interpret quantifiers such as *several*, *many*, and *few* to mean higher quantities when the speaker uses concomitant facial expressions such as brow raising/eye widening – representing larger sizes – as opposed to frowning or eye squinting – representing smaller sizes.

While it is plausible to assume that participants' interpretation of *several* was influenced by the space that was created between the speakers' hands, the influence of size may also have been one of salience: it might be that extremely small or large gestures are simply more salient which is why they influenced participants' interpretation of *several* more than moderate gestures. Looking at data of experiment I suggests that saliency may play a role. Considering participants who reported to have definitely noticed the gesture, participants noticed the gestures slightly more frequently for Speaker1 (81% out of all Speaker1 trials) who produced more extreme and therefore salient gestures than for Speaker2 (77% out of all Speaker2 trails). We do not know whether this difference is meaningful and the correlation between gesture saliency and the effect of gesture warrants further investigation.

It is not clear whether the gestures we used were more or less noteworthy than the gestures seen, for

example, in the TV News. As discussed previously, our gestures were modelled after size gestures that occur in naturally occurring numerical communication, based on the results of Woodin et al. (2020). In this data, the hands are often – just as in our task – quite elevated so that they are visible in the screen. From this perspective, our stimuli are not that unusual. However, it has to be stated that the clearly noticeable gesture effect differs from some of the other studies we discussed above (e.g. Broaders & Goldwin-Meadow, 2010; Gurney, Ellis & Vardon-Hynard, 2013), where gesture was found to exert influence without being noticed. Future research should therefore look at cases in which the gesture is less noteworthy to assess whether it has an influence even if going unnoticed.

6.2 Quantifiers: range or aspects of meaning?

Overall, the gestural effect found for quantifier comprehension is in line with previous research showing that visual information, such as the size of visually presented objects, influences the interpretation of quantifiers (Hörmann, 1983; Newstead & Coventry, 2000), as well as with the general finding that language comprehension is influenced by visual context (Degen & Tanenhaus, 2016; Spivey et al., 2002). It still remains open precisely how inward- versus outward-directed gestures influence the interpretation of *several*. We discuss two different ways in which our results could be interpreted. Firstly, gestures may simply restrict/extend the range of quantities associated with quantifiers, such that comprehenders seeing an inward gesture consider lower ranges of possible quantities than comprehenders seeing an outward gesture. Here, comprehenders would mostly rely on their association between space and magnitude to interpret *several*, which would similarly apply for any expression of quantity.

Alternatively, the effect of gesture may rely on another dimension, namely monotonicity, and the gestural effect may be unique to the interpretation of *several* and other quantifiers which are not clearly monotone, such as *a few*. *Several* is generally argued to be an upward monotone quantifier which allow inferences from a more specific to a more general scope, such as *some*, *many* and *most* (Barwise & Cooper, 1981), with the corresponding reading *at least several*. Whereas most

quantifiers are either known as upward- or downward-monotone, *several* and *a few* belong to a class that can be either upward- or non-monotone and context constrains how the quantifiers are interpreted (Barwise & Cooper, 1981). Thus, besides the upward-monotone reading *at least several*, Barwise and Cooper (1981) argued that there are instances in which *several* may in fact be described as non-monotone, thereby meaning *several but not many*, as in (4).

(4) Many students attended the pragmatics lecture and several students attended the stats lecture.

Thus, it is possible that the influence of gesture is actually rooted in modulating between the two readings of *at least several* and *several but not many*. While the outwards-directed gesture may have highlighted the upward-monotone reading *at least several*, the horizontal inwards-directed gesture may have highlighted the non-monotone reading *several but not many*. This would explain why assigned quantities in the inward gesture condition were on average lower and varied less than assigned quantities in the outward condition: the reading *several but not many* restricts the quantity range more than the open-ended reading *at least several*.

It remains to be investigated whether gestures extend/restrict the range of quantities associated with quantifiers per se, independently of their monotonicity. In such a scenario we would assume that quantities associated with, for example, *many* and *few* – arguably strictly upward or downward monotone quantifiers – would be on average higher when being presented together with an outward as opposed to an inward gesture. Alternatively, if it is about the bivalent monotonicity of quantifiers such as *several*, we would not expect the strictly upward monotone quantifier *many* or the strictly downward monotone quantifier *few* to be affected by gesture to the same extent. Besides monotonicity, there are other features distinguishing quantifiers that may play a role in whether their interpretation is susceptible to gesture, such as vagueness. In future studies we plan to test the influence of gesture on a wider range of quantifiers to shed more light on their distinct features.

6.3 Variation between contexts and individual differences

Our results showed that the interpretation of *several* very much depended on the discourse context and individual preferences. The high variation in assigned quantities between different discourse contexts is in line with previous research which postulated that comprehenders' prior expectations affect their interpretation of quantifiers (Moxey & Sanford, 1993a, 1993b). Furthermore, our data suggests individual differences between what comprehenders understand *several* to mean and whether and how strongly their interpretation is affected by gestures. While not every comprehender's interpretation of *several* was found to be influenced by gestures, we were able to demonstrate that the effect of gesture affects people reliably such that the gesture effect is not driven by a small subset of participants but instead by the majority of the participants.

Methodologically, it is worth highlighting that our statistical models do not only model the average effect as a function of gesture, but also its variation (the 'precision' parameter of beta binomial models). To clarify, our models, akin to conventional models, estimate mean response differences between gesture conditions, but they also capture the variability within these estimates. Specifically, we observed greater variability in estimates for the outward gesture condition compared to the inward or no gesture conditions, and higher variability for larger gestures (e.g., Speaker 1 in the outward condition) than for smaller gestures (e.g., Speaker 2 in the outward condition). Our findings show that it is important to not only look at whether gesture influences averages, but also whether different gestures lead to more or less variation, as demonstrated here.

To sum up, based on our overall results, a model capturing the interpretation of *several* should be able to account for the speakers' manual gestures, the discourse and communicative context, and the comprehenders' individual preferences regarding what *several* means and whether or not – and to what extent – they consider manual gestures in their interpretation of *several*.

6.4 Outlook

This investigation makes contributions to several distinct literatures. Overall, our study has implications for research on the processing of numerical information, as has been done by Andres, Davare, Pesenti, Olivier and Seron (2004); Andres, Ostry, Florence and Paus (2008); Badets, Bouquet, Ric and Pesenti (2012); Grade, Badets and Presenti (2017); Lindemann, Abolafia, Girardi and Bekkering (2007). In contrast to existing numerical cognition research, which often conducts studies of how people comprehend isolated stimuli such as numerals, one advantage of our approach is its ecological validity. Our stimuli were modelled after real uses of quantifiers, and our gestures after real uses of gestures of magnitude (Woodin et al., 2020). This allowed us to test spatial-numerical associations in a way that shows its role in real world communication. Another constraint of existing lab-based research is that the investigated number ranges are usually visually presented numerals 1-9 (Dehaene, Bossini & Giraux, 1993; Wood, Willmes, Nuerk & Fischer, 2008), but quantifiers – precisely because they do not encode precise numerical values but rather ranges of values – make it possible to investigate the connection between space and number for a much larger range of quantities. In this sense, our study dovetails with the observational research on gesture production in numerical research (Woodin et al., 2020; Carrion & Valenzuela, 2021), in that we have shown that the gestures which frequently feature in numerical communication actually make a difference in understanding.

Furthermore, by investigating whether quantifier comprehension is influenced by gesture, our study serves as a case study for another dimension of contextual influence in quantifier comprehension which is relevant for the areas of semantics and pragmatics. Our research moreover extends the contexts in which the influence of gesture on comprehension has been studied so far, such as Nicol and Patson's (2022) study on plural conceptualisation but also gestural research on the temporal domain in which gestures were found to be able to resolve the ambiguity of a message. For example, when asked the question "Next Wednesday's meeting is moved forward two days, what day is it on now?", about half of all people respond with Friday, the other half with Monday (McGlone

& Harding, 1998; Stickles & Lewis, 2018). This ambiguity is resolved when people see somebody gesturing forwards or backwards while asking the same question such that people most frequently respond with Friday or Monday respectively (Jamalian & Tversky, 2012; Lewis & Stickles, 2017; Winter & Duffy, 2020).

Lastly, our findings have implications for sign language research studying ways in which imprecise quantities are expressed through size-related signs. Previous research already established connections between sign languages and spoken languages in terms of how signs and gesture encode set-sizes and relations. More specifically, Durkin, Gagne and Davidson's (2016) research demonstrated that American Sign Language (ASL) signers and American English speakers both express sizes with size-related signs and gestures: ASL signers were found to use signs on the vertical axis to express set sizes and subset/superset relationships, i.e. signs located higher express larger set sizes and supersets, and American English speakers used vertical gestures and a combination of vertical and horizontal gestures to indicate set sizes and subset/superset relationships.

7 Conclusion

This study is the first to demonstrate that quantifier interpretation is susceptible to gesture and that gesture can influence what numbers comprehenders have in mind. Our findings provide evidence that numerical estimates can be shifted in line with spatial-numerical associations triggered by gestural contexts in alignment with numerical cognition studies. These results indicate that English speakers may use quantifiers in conjunction with outwards- and inwards-directed gestures to frame quantities as smaller or larger. Such a strategy is particularly persuasive, since the production of quantifiers escapes the scrutiny that the production of exact numbers can face.

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8 Appendix I

8.1 Experimental Items

Number	Domain	Stimuli
1	Politics	400 voters were interviewed. Several of them reported to have voted for labour.
2	Politics	400 people were at the protest. Several of them got arrested.
3	Science	400 people participated in the trial. Several of them experienced side effects.
4	Science	400 new research projects received funding. Several of them are in Humanities.
5	Industry	400 cars were produced that day. Several of them were luxury cars.
6	Industry	400 laptops were imported today. Several of them were damaged.
7	Climate change	400 climate events occurred last year. Several of them were catastrophic.
8	Climate change	400 new trees were planted this quarter. Several of them were birches.
9	Insurance	400 people got car insurance this month. Several of them insured brand new cars.
10	Insurance	400 insurance claims were filed after the event. Several of them were rejected.
11	Immigration	400 asylum seekers were granted refugee status. Several of them were young children.
12	Immigration	400 migrants made the crossing last year. Several of them were sent back.
13	Accidents	400 road accidents were reported this month. Several of them involved fatalities.
14	Accidents	400 people were on the cruise that sank. Several of them sustained critical injuries.
15	Building projects	400 historical buildings are currently derelict. Several of them are set to be restored.
16	Building projects	400 new homes are being built under the scheme. Several of them were already sold.
17	Office	400 monitors are documented in the inventory. Several of them are used by IT.
18	Office	400 new files were ordered. Several of them were ordered by accounting.
19	Animal	400 elephants were born this month. Several of them were documented in India.
20	Animal	400 new insect species were discovered this year. Several of them were beetles.

Table 5: Experimental Stimuli Experiment I

9 Appendix II

9.1 Plots: experiment I

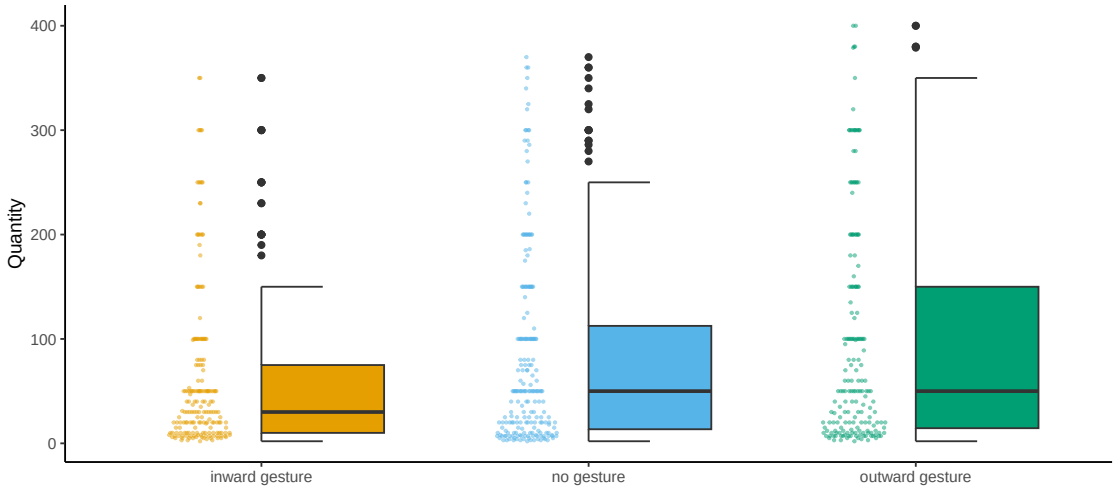


Figure 9: Raw (dot plot) and median by-gesture quantities (box plot).

9.2 Model 1

Coefficient	posterior mean	Standard Error	l-95% CrI	u-95% CrI
μ intercept	-1.26	0.09	-1.44	-1.07
μ inward gesture	-0.36	0.13	-0.61	-0.11
μ outward gesture	0.30	0.13	0.05	0.55
ϕ intercept	1.06	0.10	0.86	1.24
ϕ inward gesture	0.40	0.15	0.12	0.69
ϕ outward gesture	-0.33	0.14	-0.60	-0.06

Table 6: Population-level estimates of the Bayesian beta binomial regression model with 95% credible intervals. The μ coefficients are on the log-odds scale. The intercept is the predicted log-odds for the no gesture condition. The categorical predictor gesture was treatment-coded with no gesture as reference level. The ϕ parameters are on the log scale. The intercept is the predicted precision for the distribution over quantities in the no gesture condition and the ϕ inward/outward gesture coefficients are the changes in precision of the distribution over quantities when changing to the inward or outward gesture condition.

9.2.1 Posterior distributions over population-level estimates

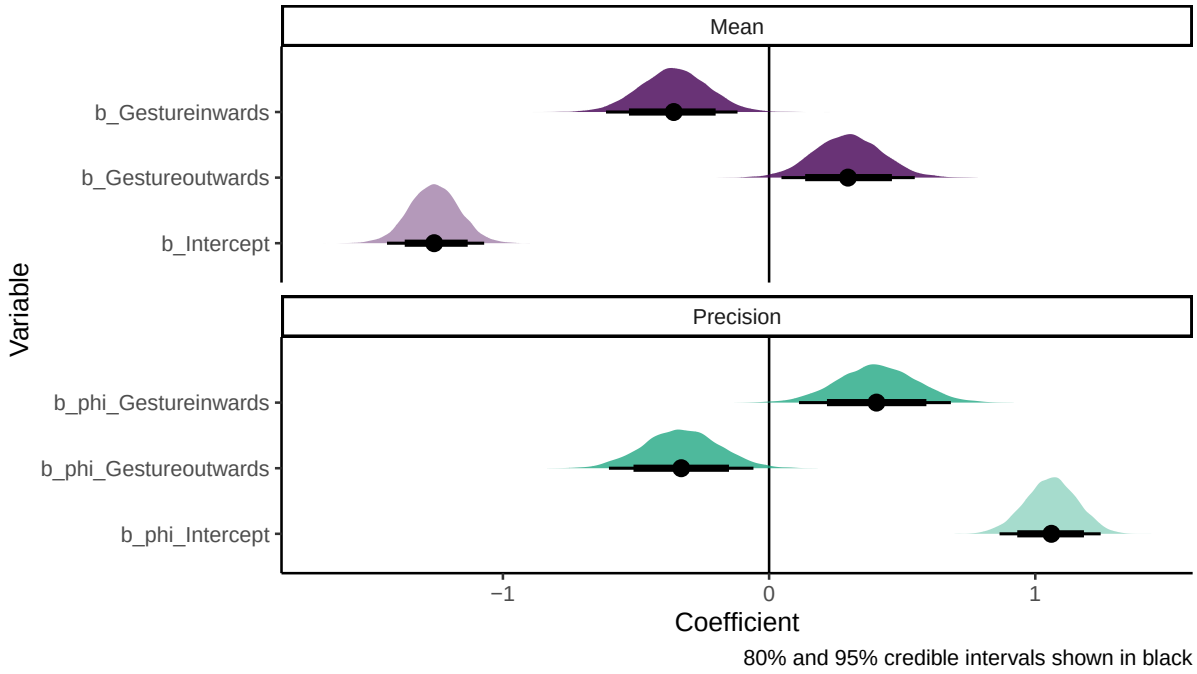


Figure 10: Posterior distribution over population-level estimates with 80% and 95% credible intervals.

9.2.2 Random effects estimates

Coefficient	posterior mean	Standard Error	l-95% CrI	u-95% CrI
Item (levels: 20)				
sd(Intercept)	0.15	0.08	0.01	0.32
sd(Gestureinward)	0.16	0.11	0.01	0.43
sd(Gestureoutward)	0.11	0.09	0.00	0.32
cor(Intercept, Gestureinward)	0.05	0.40	-0.72	0.77
cor(Intercept, Gestureoutward)	-0.08	0.41	-0.79	0.70
cor(Gestureinward, Gestureoutward)	0.03	0.41	-0.75	0.77
Speaker2:Item (levels: 40)				
sd(Intercept)	0.09	0.06	0.00	0.24
sd(Gestureinward)	0.12	0.09	0.00	0.32
sd(Gestureoutward)	0.12	0.09	0.00	0.32
cor(Intercept,Gestureinward)	-0.03	0.41	-0.77	0.74
cor(Intercept,Gestureoutward)	-0.10	0.41	-0.81	0.72
cor(Gestureinward,Gestureoutward)	0.00	0.41	-0.75	0.77

Table 7: Random-effects of the Gesture Bayesian beta binomial regression model with 95% credible intervals.

9.3 Model 1 (gesture as a binary factor)

Coefficient	posterior mean	Standard Error	l-95% CrI	u-95% CrI
μ intercept	-1.60	0.10	-1.79	-1.41
μ outward gesture	0.63	0.13	0.37	0.90
ϕ intercept	1.43	0.11	1.20	1.65
ϕ outward gesture	-0.70	0.15	-0.99	-0.41

Table 8: Population-level estimates of the Bayesian beta binomial regression model with 95% credible intervals for experiment II. The μ coefficients are on the log-odds scale. The intercept is the predicted log-odds for the inward gesture condition. The categorical predictor gesture was treatment-coded with the inward gestures as reference level the no gesture condition was excluded. The ϕ parameters are on the log scale. The intercept is the predicted precision for the distribution over quantities in the inward gesture condition and the ϕ outward gesture coefficient is the change in precision of the distribution over quantities when changing to the outward gesture condition.

9.3.1 Posterior distributions over population-level estimates

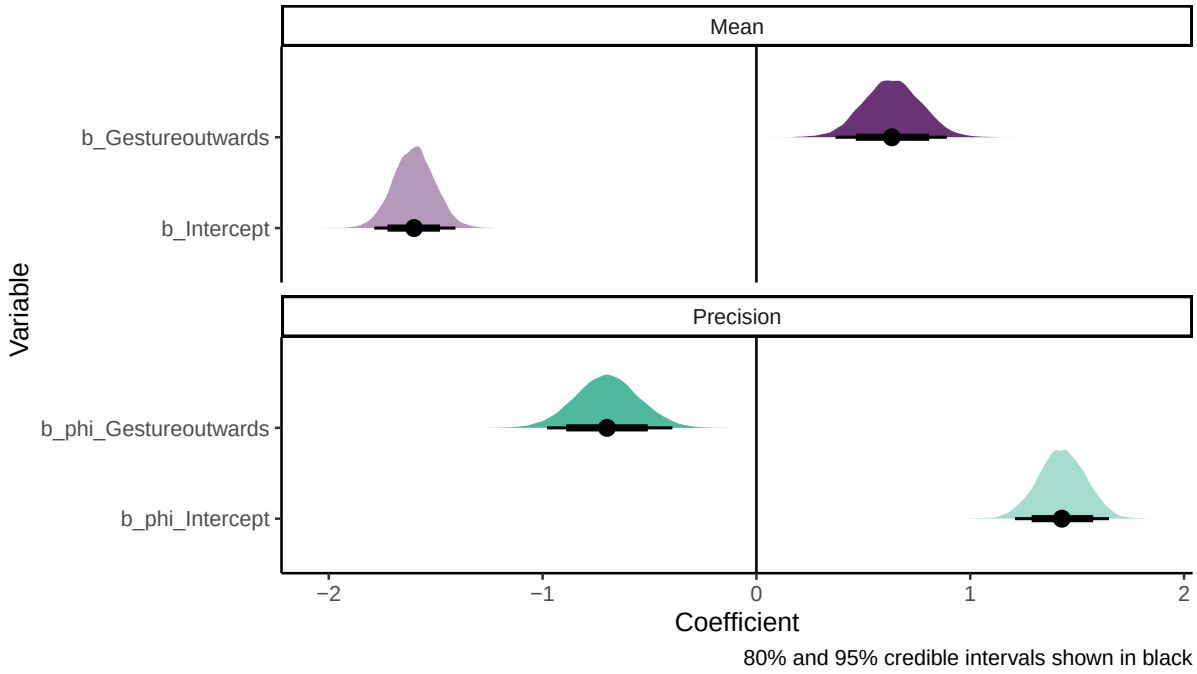


Figure 11: Posterior distribution over population-level estimates with 80% and 95% credible intervals.

9.4 Random effects estimates

Coefficient	posterior mean	Standard Error	l-95% CrI	u-95% CrI
Item (levels: 20)				
sd(Intercept)	0.15	0.10	0.01	0.37
sd(Gestureoutward)	0.12	0.09	0.00	0.35
cor(Intercept,Gestureoutward)	-0.13	0.45	-0.88	0.77
Speaker:Item (levels: 40)				
sd(Intercept)	0.09	0.07	0.00	0.26
sd(Gestureoutward)	0.12	0.09	0.01	0.34
cor(Intercept,Gestureoutward)	-0.11	0.45	-0.86	0.76

Table 9: Random-effects of the Gesture Bayesian beta binomial regression model (Model A1 binary) with 95% credible intervals.

9.5 Model 2

Coefficient	posterior mean	Standard Error	l-95% CrI	u-95% CrI
μ intercept	-1.27	0.07	-1.41	-1.14
μ gesture1	0.34	0.08	0.19	0.49
μ gesture2	-0.35	0.07	-0.49	-0.20
μ speaker1	-0.05	0.05	-0.16	0.05
μ gesture1:speaker1	0.13	0.07	-0.01	0.27
μ gesture2:speaker1	-0.11	0.06	-0.24	0.01
ϕ intercept	1.10	0.06	0.98	1.21
ϕ gesture1	-0.37	0.08	-0.53	-0.20
ϕ gesture2	0.39	0.09	0.21	0.55
ϕ speaker1	0.12	0.06	0.01	0.24

Table 10: Population-level estimates of the Bayesian beta binomial regression model with 95% credible intervals. The μ coefficients are on the log-odds scale. The coefficient gesture1 (0.34) is the difference between grand intercept and the outward condition, the coefficient gesture2 (-0.35) is the difference between intercept and the inward condition. The difference between intercept and no gesture condition can be calculated as follows $= 0.34 * (-1) - 0.35 * (-1) = -0.01$. The ϕ parameters are on the log scale. The intercept is the predicted precision for the distribution over quantities averaged over conditions and speakers. The ϕ gesture1/speaker1 coefficients are the changes in precision of the distributions over quantities when changing to the outward/inward gesture condition/speaker1.

9.5.1 Contrast coding

	gesture1	gesture2
outward	1	0
inward	0	1
no	-1	-1

Table 11: ModelA2: Gesture sum-coded.

9.5.2 Posterior distributions over population-level estimates

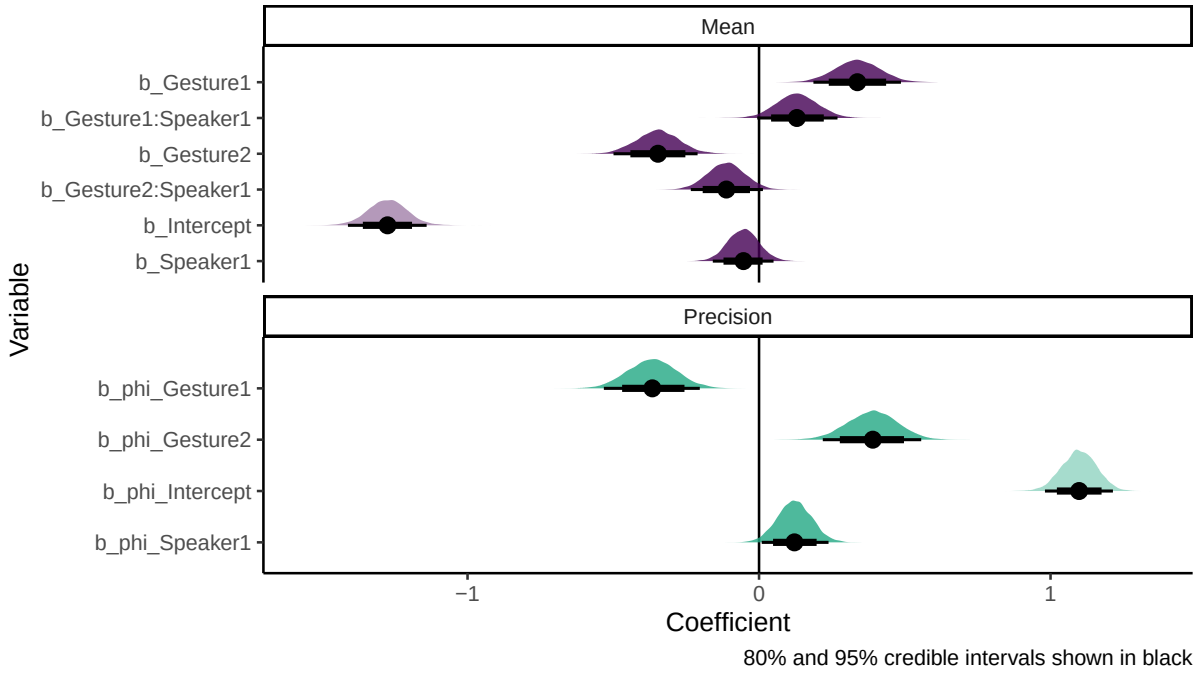


Figure 12: Posterior distribution over population-level estimates with 80% and 95% credible intervals.

9.6 Random effects estimates

Coefficient	posterior mean	Standard Error	l-95% CrI	u-95% CrI
Item (levels: 20)				
sd(Intercept)	0.19	0.08	0.03	0.35
sd(Gesture1)	0.06	0.05	0.00	0.19
sd(Gesture2)	0.07	0.06	0.00	0.20
sd(Speaker1)	0.06	0.05	0.00	0.17
sd(Gesture1:Speaker1)	0.09	0.07	0.00	0.25
sd(Gesture2:Speaker1)	0.06	0.05	0.00	0.18
cor(Intercept,Gesture1)	-0.03	0.33	-0.64	0.62
cor(Intercept,Gesture2)	0.05	0.33	-0.59	0.66
cor(Gesture1,Gesture2)	-0.05	0.34	-0.67	0.60
cor(Intercept,Speaker1)	-0.01	0.33	-0.64	0.63
cor(Gesture1,Speaker1)	0.01	0.34	-0.63	0.65
cor(Gesture2,Speaker1)	0.00	0.33	-0.62	0.62
cor(Intercept,Gesture1:Speaker1)	-0.00	0.32	-0.61	0.61
cor(Gesture1,Gesture1:Speaker1)	-0.00	0.34	-0.64	0.64
cor(Gesture2,Gesture1:Speaker1)	-0.01	0.33	-0.64	0.62
cor(Speaker1,Gesture1:Speaker1)	-0.02	0.33	-0.64	0.62
cor(Intercept,Gesture2:Speaker1)	-0.04	0.33	-0.65	0.60
cor(Gesture1,Gesture2:Speaker1)	0.00	0.33	-0.63	0.62
cor(Gesture2,Gesture2:Speaker1)	-0.02	0.34	-0.65	0.62
cor(Speaker1,Gesture2:Speaker1)	-0.01	0.33	-0.63	0.61
cor(Gesture1:Speaker1,Gesture2:Speaker1)	-0.04	0.33	-0.65	0.61

Table 12: Random-effects of the Speaker Bayesian beta binomial regression model with 95% credible intervals.

9.7 Model 3

9.7.1 Posterior distributions over population-level estimates

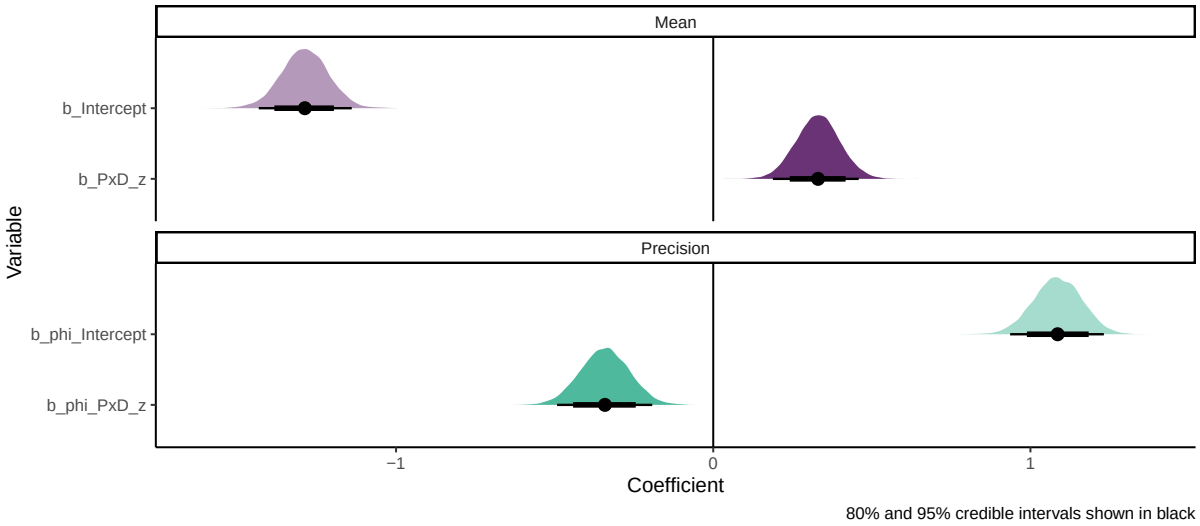


Figure 13: Posterior distribution over population-level estimates with 80% and 95% credible intervals for Model 3.

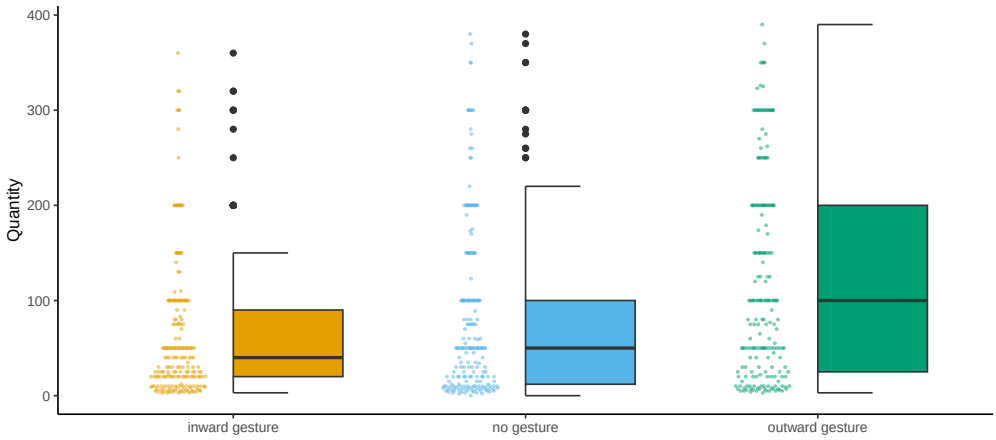
9.8 Random effects estimates

Coefficient	posterior mean	Standard Error	l-95% CrI	u-95% CrI
Item (levels: 20)				
sd(Intercept)	0.16	0.09	0.01	0.35
sd(PxD$_z$)	0.07	0.06	0.00	0.22
cor(Intercept, PxD$_z$)	-0.09	0.44	-0.84	0.77

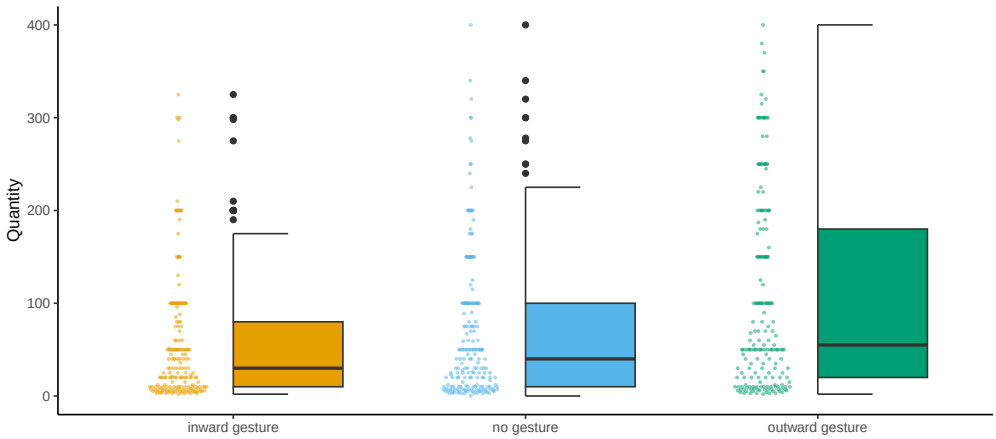
Table 13: Random-effects of the pixel distance Bayesian beta binomial regression model with 95% credible intervals (Model 3).

10 Appendix III

10.1 Plots: experiments II and III



(a) Experiment II



(b) Experiment III

Figure 14: Raw (dot plot) and median by-gesture quantities (box plot)

10.2 Model 1: experiment II

Coefficient	posterior mean	Standard Error	l-95% CrI	u-95% CrI
μ intercept	-1.50	0.11	-1.71	-1.29
μ inward gesture	-0.41	0.10	-0.61	-0.22
μ outward gesture	0.58	0.11	0.37	0.79
ϕ intercept	2.01	0.14	1.74	2.28
ϕ inward gesture	0.61	0.22	0.18	1.07
ϕ outward gesture	-0.50	0.17	-0.84	-0.17

Table 14: Population-level estimates of the Bayesian beta binomial regression model with 95% credible intervals for experiment II. The μ coefficients are on the log-odds scale. The intercept is the predicted log-odds for the no gesture condition. The categorical predictor gesture was treatment-coded with no gesture as reference level. The ϕ parameters are on the log scale. The intercept is the predicted precision for the distribution over quantities in the no gesture condition and the ϕ inward/outward gesture coefficients are the changes in precision of the distribution over quantities when changing to the inward or outward gesture condition.

10.2.1 Posterior distributions over population-level estimates

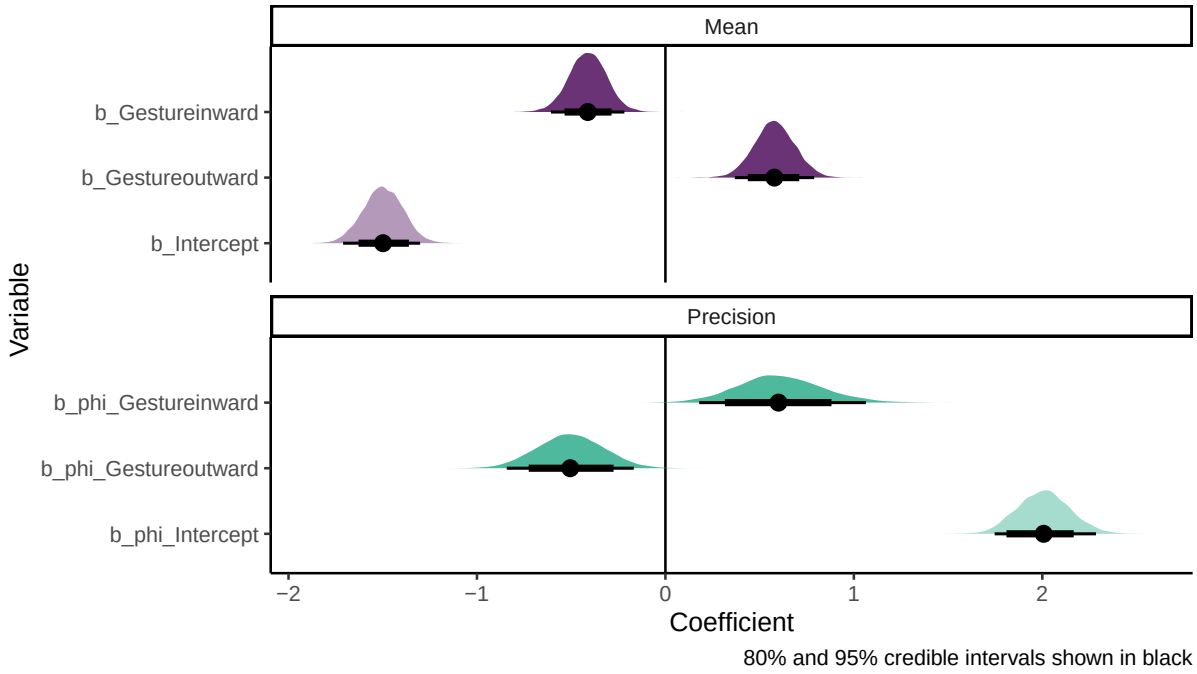


Figure 15: Posterior distribution over population-level estimates with 80% and 95% credible intervals.

10.2.2 Random effects estimates

Coefficient	posterior mean	Standard Error	l-95% CrI	u-95% CrI
Item (levels: 20)				
sd(Intercept)	0.16	0.09	0.01	0.34
sd(Gestureinward)	0.10	0.08	0.00	0.30
sd(Gestureoutward)	0.14	0.10	0.01	0.38
cor(Intercept,Gestureinward)	-0.10	0.41	-0.80	0.71
cor(Intercept,Gestureoutward)	-0.11	0.41	-0.81	0.69
cor(Gestureinward,Gestureoutward)	0.05	0.41	-0.73	0.77
Speaker2:Item (levels: 40)				
sd(Intercept)	0.15	0.09	0.01	0.35
sd(Gestureinward)	0.22	0.13	0.01	0.50
sd(Gestureoutward)	0.20	0.13	0.01	0.48
cor(Intercept,Gestureinward)	-0.19	0.40	-0.83	0.66
cor(Intercept,Gestureoutward)	-0.01	0.39	-0.74	0.73
cor(Gestureinward,Gestureoutward)	-0.01	0.39	-0.73	0.73

Table 15: Random-effects of the Gesture Bayesian beta binomial regression model with 95% credible intervals.

10.3 Model 1 (binary): experiment II

Coefficient	posterior mean	Standard Error	l-95% CrI	u-95% CrI
μ intercept	-1.62	0.09	-1.79	-1.45
μ outward gesture	0.79	0.12	0.54	1.03
ϕ intercept	1.53	0.11	1.31	1.74
ϕ outward gesture	-0.62	0.14	-0.90	-0.34

Table 16: Population-level estimates of the Bayesian beta binomial regression model with 95% credible intervals for experiment II. The μ coefficients are on the log-odds scale. The intercept is the predicted log-odds for the inward gesture condition. The categorical predictor gesture was treatment-coded with the inward gestures as reference level the no gesture condition was excluded. The ϕ parameters are on the log scale. The intercept is the predicted precision for the distribution over quantities in the inward gesture condition and the ϕ outward gesture coefficient is the change in precision of the distribution over quantities when changing to the outward gesture condition.

10.3.1 Posterior distributions over population-level estimates

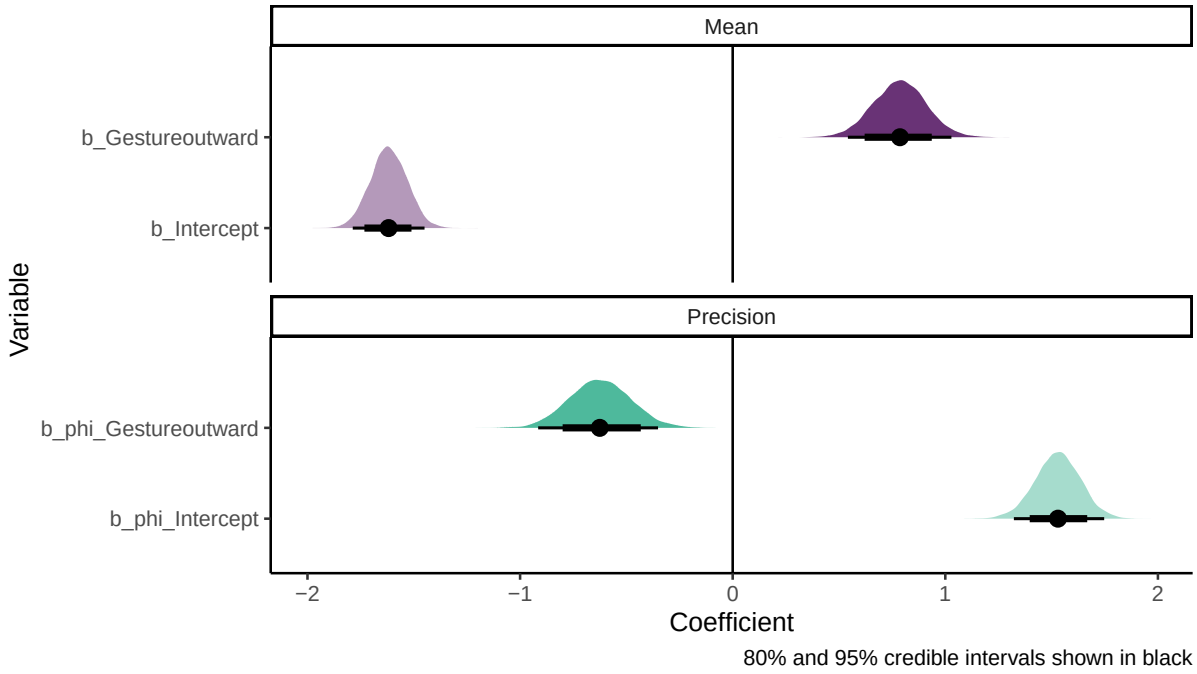


Figure 16: Posterior distribution over population-level estimates with 80% and 95% credible intervals.

10.4 Random effects estimates

Coefficient	posterior mean	Standard Error	l-95% CrI	u-95% CrI
Item (levels: 20)				
sd(Intercept)	0.09	0.07	0.00	0.25
sd(Gestureoutward)	0.13	0.10	0.00	0.36
cor(Intercept,Gestureoutward)	-0.08	0.45	-0.84	0.79
Speaker:Item (levels: 40)				
sd(Intercept)	0.09	0.07	0.00	0.26
sd(Gestureoutward)	0.22	0.14	0.01	0.52
cor(Intercept,Gestureoutward)	-0.12	0.45	-0.87	0.76

Table 17: Random-effects of the Gesture Bayesian beta binomial regression model (Model A1 binary) with 95% credible intervals.

10.5 Model 2: experiment II

Coefficient	posterior mean	Standard Error	l-95% CrI	u-95% CrI
μ intercept	-1.46	0.09	-1.63	-1.29
μ gesture1	0.52	0.06	0.40	0.65
μ gesture2	-0.45	0.05	-0.56	-0.35
μ speaker1	0.12	0.08	-0.03	0.28
μ gesture1:speaker1	0.15	0.06	0.04	0.26
μ gesture2:speaker1	-0.13	0.07	-0.26	0.02
ϕ intercept	2.10	0.09	1.92	2.28
ϕ gesture1	-0.55	0.11	-0.77	-0.33
ϕ gesture2	0.55	0.14	0.30	0.83
ϕ speaker1	-0.15	0.08	-0.30	0.00

Table 18: Population-level estimates of the Bayesian beta binomial regression model with 95% credible intervals. The μ coefficients are on the log-odds scale. The coefficient gesture1 (0.52) is the difference between grand intercept and the outward condition, the coefficient gesture2 (-0.45) is the difference between intercept and the inward condition. The difference between intercept and no gesture condition can be calculated as follows $= 0.52 * (-1) - 0.45 * (-1) = -0.07$. The ϕ parameters are on the log scale. The intercept is the predicted precision for the distribution over quantities averaged over conditions and speakers. The ϕ gesture1/speaker1 coefficients are the changes in precision of the distributions over quantities when changing to the outward/inward gesture condition/speaker1.

10.5.1 Posterior distributions over population-level estimates

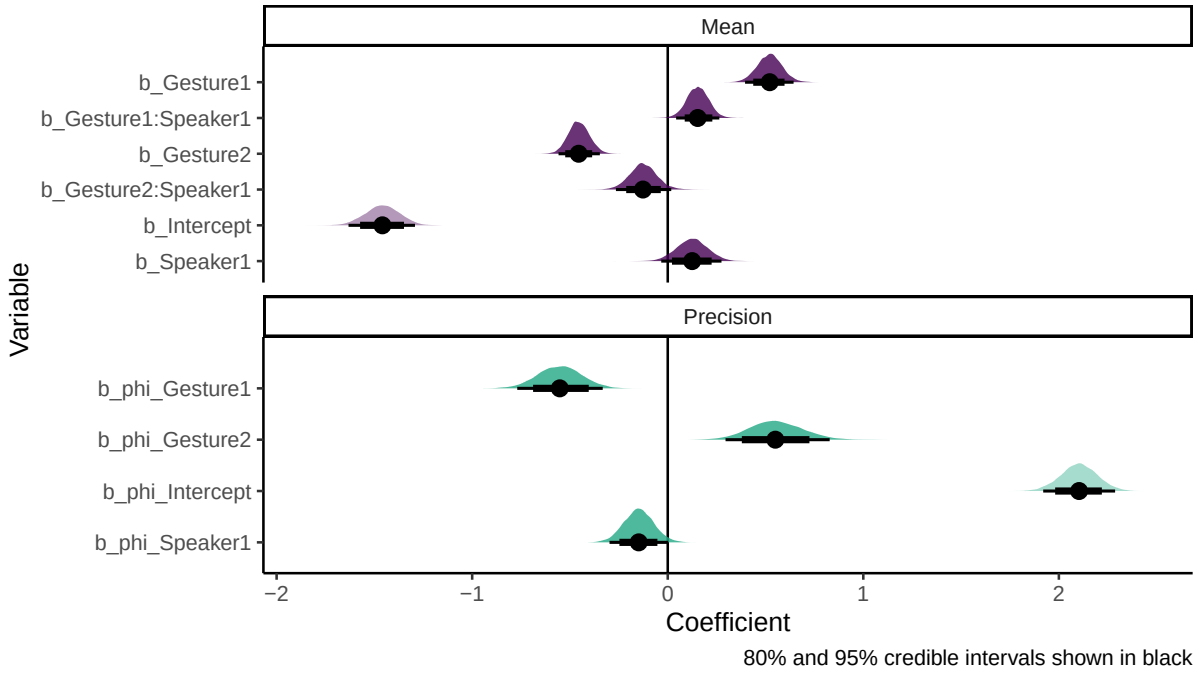


Figure 17: Posterior distribution over population-level estimates with 80% and 95% credible intervals.

10.6 Random effects estimates

Coefficient	posterior mean	Standard Error	l-95% CrI	u-95% CrI
Item (levels: 20)				
sd(Intercept)	0.20	0.07	0.06	0.34
sd(Gesture1)	0.11	0.07	0.01	0.28
sd(Gesture2)	0.08	0.06	0.00	0.22
sd(Speaker1)	0.11	0.07	0.01	0.26
sd(Gesture1:Speaker1)	0.09	0.07	0.00	0.24
sd(Gesture2:Speaker1)	0.23	0.09	0.06	0.41
cor(Intercept,Gesture1)	-0.11	0.32	-0.68	0.53
cor(Intercept,Gesture2)	-0.10	0.32	-0.68	0.54
cor(Gesture1,Gesture2)	-0.00	0.33	-0.63	0.63
cor(Intercept,Speaker1)	0.14	0.31	-0.50	0.69
cor(Gesture1,Speaker1)	0.01	0.33	-0.61	0.63
cor(Gesture2,Speaker1)	-0.01	0.33	-0.64	0.62
cor(Intercept,Gesture1:Speaker1)	0.06	0.33	-0.58	0.67
cor(Gesture1,Gesture1:Speaker1)	-0.01	0.33	-0.64	0.62
cor(Gesture2,Gesture1:Speaker1)	-0.01	0.33	-0.63	0.63
cor(Speaker1,Gesture1:Speaker1)	0.05	0.33	-0.59	0.65
cor(Intercept,Gesture2:Speaker1)	0.08	0.29	-0.48	0.62
cor(Gesture1,Gesture2:Speaker1)	-0.15	0.32	-0.71	0.51
cor(Gesture2,Gesture2:Speaker1)	-0.12	0.33	-0.71	0.55
cor(Speaker1,Gesture2:Speaker1)	0.01	0.30	-0.58	0.59
cor(Gesture1:Speaker1,Gesture2:Speaker1)	-0.05	0.33	-0.66	0.60

Table 19: Random-effects of the Speaker Bayesian beta binomial regression model with 95% credible intervals.

10.7 Model 3: experiment II

Coefficient	posterior mean	Standard Error	l-95% CrI	u-95% CrI
μ intercept	-1.37	0.08	-1.53	-1.22
μ distance (px)	0.49	0.06	0.37	0.61
ϕ intercept	1.86	0.13	1.60	2.11
ϕ distance (px)	-0.44	0.10	-0.65	-0.25

Table 20: Population-level estimates of the Bayesian beta binomial regression model on the log-odds scale with 95% credible intervals. The continuous predictor Distance (px) was standardised.

10.7.1 Posterior distributions over population-level estimates

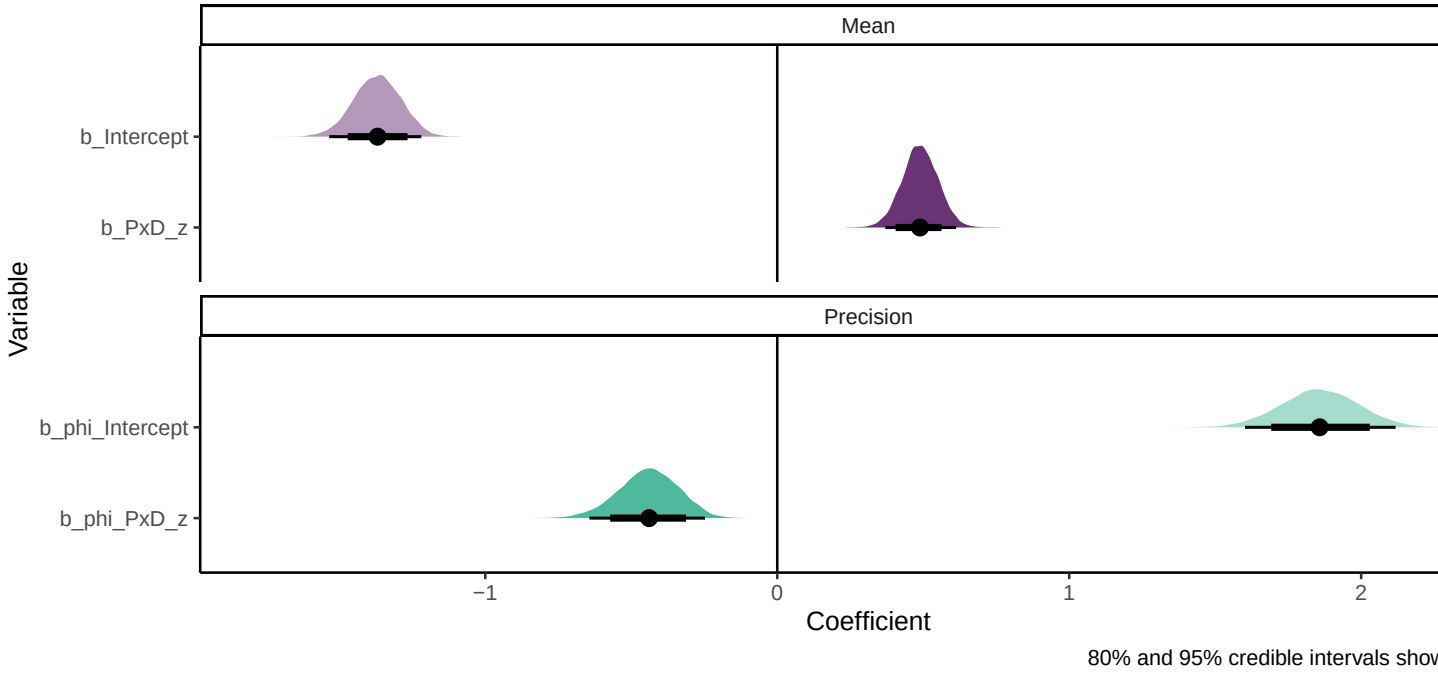


Figure 18: Posterior distribution over population-level estimates with 80% and 95% credible intervals.

10.8 Random effects estimates

Coefficient	posterior mean	Standard Error	l-95% CrI	u-95% CrI
Item (Number of levels: 20)				
sd(Intercept)	0.10	0.07	0.00	0.26
sd(PxD_z)	0.11	0.07	0.00	0.27
cor(Intercept,PxD_z)	0.01	0.44	-0.80	0.81
Participant (Number of levels: 193)				
sd(Intercept)	0.78	0.09	0.60	0.95

Table 21: Random-effects of the Pixel Distance Bayesian beta binomial regression model with 95% credible intervals.

10.9 Model 1: experiment III

Coefficient	posterior mean	Standard Error	l-95% CrI	u-95% CrI
μ Intercept	-1.80	0.12	-2.04	-1.57
μ inward gesture	-0.37	0.14	-0.63	-0.10
μ outward gesture	0.66	0.11	0.43	0.88
ϕ intercept	2.46	0.18	2.13	2.82
ϕ inward gesture	0.47	0.25	-0.03	0.97
ϕ outward gesture	-0.82	0.20	-1.21	-0.43

Table 22: Population-level estimates of the Bayesian beta binomial regression model with 95% credible intervals for experiment III. The μ coefficients are on the log-odds scale. The intercept is the predicted log-odds for the no gesture condition. The categorical predictor gesture was treatment-coded with no gesture as reference level. The ϕ parameters are on the log scale. The intercept is the predicted precision for the distribution over quantities in the no gesture condition and the ϕ inward/outward gesture coefficients are the changes in precision of the distribution over quantities when changing to the inward or outward gesture condition.

10.9.1 Posterior distributions over population-level estimates

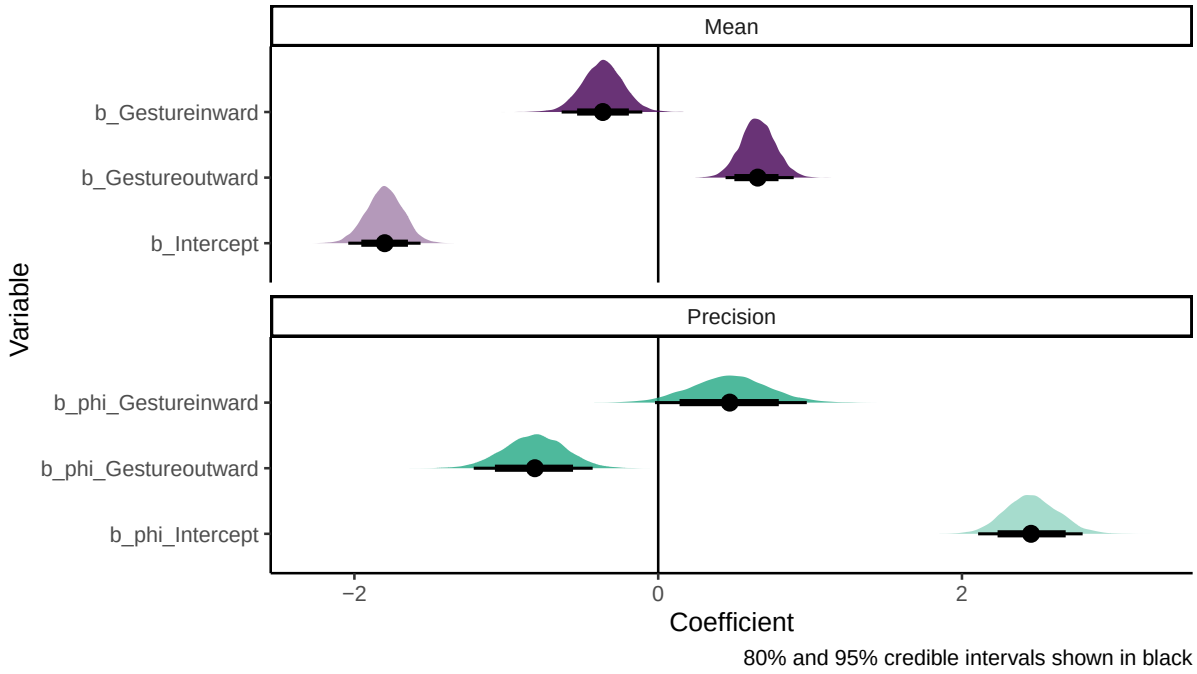


Figure 19: Posterior distribution over population-level estimates with 80% and 95% credible intervals.

10.9.2 Random effects estimates

Coefficient	posterior mean	Standard Error	l-95% CrI	u-95% CrI
Item (levels: 20)				
sd(Intercept)	0.26	0.11	0.04	0.48
sd(Gestureinward)	0.41	0.16	0.07	0.73
sd(Gestureoutward)	0.15	0.12	0.01	0.43
cor(Intercept,Gestureinward)	-0.05	0.35	-0.67	0.65
cor(Intercept,Gestureoutward)	-0.15	0.41	-0.83	0.67
cor(Gestureinward,Gestureoutward)	0.07	0.40	-0.72	0.77
Speaker2:Item (levels: 40)				
sd(Intercept)	0.15	0.10	0.01	0.37
sd(Gestureinward)	0.21	0.14	0.01	0.53
sd(Gestureoutward)	0.23	0.15	0.01	0.53
cor(Intercept,Gestureinward)	-0.16	0.41	-0.84	0.68
cor(Intercept,Gestureoutward)	-0.24	0.41	-0.87	0.64
cor(Gestureinward,Gestureoutward)	0.03	0.40	-0.73	0.76

Table 23: Random-effects of the Gesture Bayesian beta binomial regression model with 95% credible intervals.

10.10 Model 1 (binary): experiment III

Coefficient	posterior mean	Standard Error	l-95% CrI	u-95% CrI
μ intercept	-1.78	0.09	-1.96	-1.60
μ outward gesture	0.80	0.13	0.54	1.06
ϕ intercept	1.59	0.11	1.37	1.81
ϕ outward gesture	-0.75	0.15	-1.04	-0.45

Table 24: Population-level estimates of the Bayesian beta binomial regression model with 95% credible intervals for experiment II. The μ coefficients are on the log-odds scale. The intercept is the predicted log-odds for the inward gesture condition. The categorical predictor gesture was treatment-coded with the inward gestures as reference level the no gesture condition was excluded. The ϕ parameters are on the log scale. The intercept is the predicted precision for the distribution over quantities in the inward gesture condition and the ϕ outward gesture coefficient is the change in precision of the distribution over quantities when changing to the outward gesture condition.

10.10.1 Posterior distributions over population-level estimates

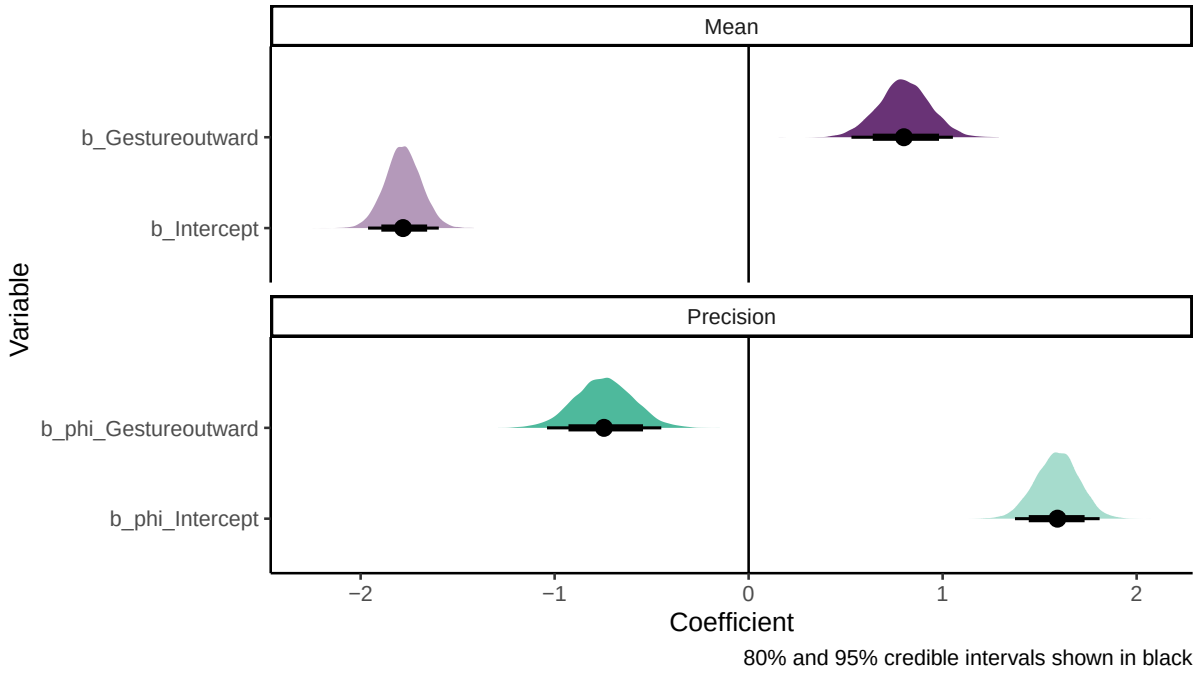


Figure 20: Posterior distribution over population-level estimates with 80% and 95% credible intervals.

10.11 Random effects estimates

Coefficient	posterior mean	Standard Error	l-95% CrI	u-95% CrI
Item (levels: 20)				
sd(Intercept)	0.12	0.09	0.01	0.32
sd(Gestureoutward)	0.18	0.12	0.01	0.46
cor(Intercept,Gestureoutward)	-0.14	0.44	-0.86	0.74
Speaker:Item (levels: 40)				
sd(Intercept)	0.08	0.06	0.00	0.24
sd(Gestureoutward)	0.14	0.10	0.01	0.39
cor(Intercept,Gestureoutward)	-0.12	0.46	-0.87	0.76

Table 25: Random-effects of the Gesture Bayesian beta binomial regression model (Model A1 binary) with 95% credible intervals.

10.12 Model 2: experiment III

Coefficient	posterior mean	Standard Error	l-95% CrI	u-95% CrI
μ intercept	-1.73	0.11	-1.96	-1.52
μ gesture1	0.57	0.06	0.45	0.70
μ gesture2	-0.45	0.08	-0.62	-0.29
μ speaker1	-0.00	0.04	-0.09	0.09
μ gesture1:speaker1	0.11	0.08	-0.05	0.27
μ gesture2:speaker1	-0.20	0.07	-0.33	-0.07
ϕ intercept	2.42	0.10	2.23	2.61
ϕ gesture1	-0.75	0.11	-0.97	-0.53
ϕ gesture2	0.57	0.15	0.29	0.88
ϕ speaker1	-0.06	0.09	-0.23	0.11

Table 26: Population-level estimates of the Bayesian beta binomial regression model with 95% credible intervals. The μ coefficients are on the log-odds scale. The coefficient gesture1 (0.34) is the difference between grand intercept and the outward condition, the coefficient gesture2 (-0.35) is the difference between intercept and the inward condition. The difference between intercept and no gesture condition can be calculated as follows $= 0.57 * (-1) - 0.45 * (-1) = -0.12$. The ϕ parameters are on the log scale. The intercept is the predicted precision for the distribution over quantities averaged over conditions and speakers. The ϕ gesture1/speaker1 coefficients are the changes in precision of the distributions over quantities when changing to the outward/inward gesture condition/speaker1.

10.12.1 Posterior distributions over population-level estimates

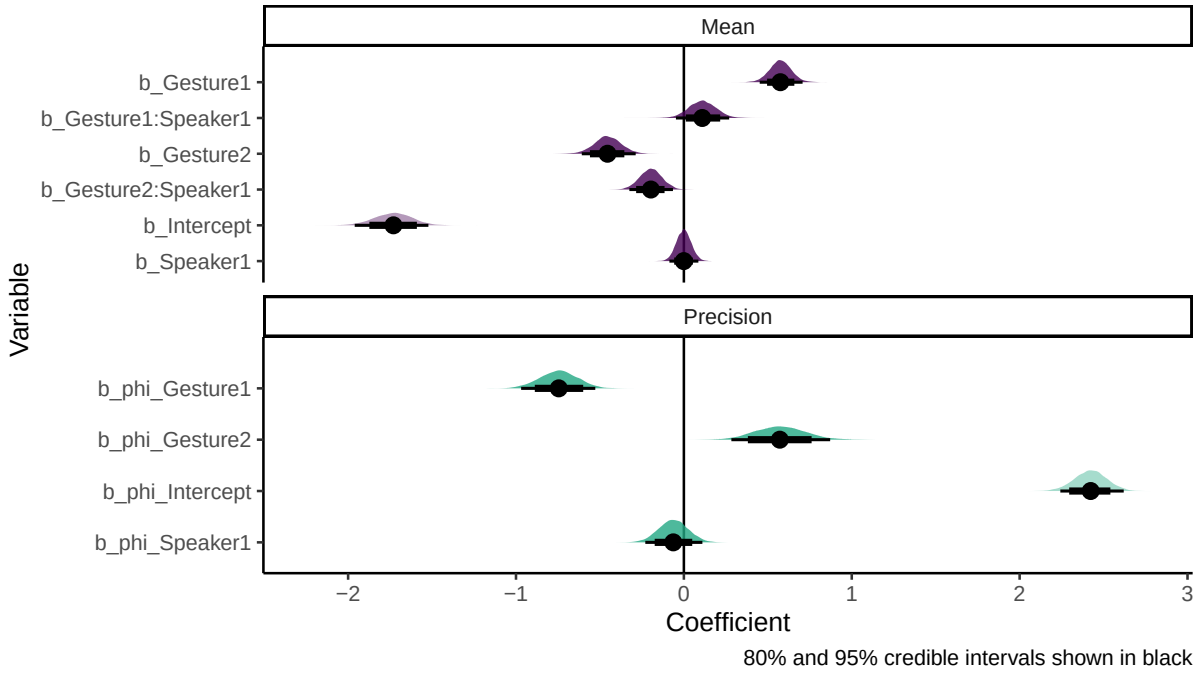


Figure 21: Posterior distribution over population-level estimates with 80% and 95% credible intervals.

10.13 Random effects estimates

Coefficient	posterior mean	Standard Error	l-95% CrI	u-95% CrI
Item (levels: 20)				
sd(Intercept)	0.33	0.08	0.20	0.50
sd(Gesture1)	0.13	0.08	0.01	0.32
sd(Gesture2)	0.28	0.09	0.13	0.47
sd(Speaker1)	0.08	0.05	0.00	0.20
sd(Gesture1:Speaker1)	0.22	0.09	0.05	0.40
sd(Gesture2:Speaker1)	0.08	0.06	0.00	0.22
cor(Intercept,Gesture1)	-0.16	0.30	-0.69	0.46
cor(Intercept,Gesture2)	0.15	0.25	-0.34	0.60
cor(Gesture1,Gesture2)	-0.16	0.31	-0.70	0.49
cor(Intercept,Speaker1)	-0.15	0.31	-0.70	0.50
cor(Gesture1,Speaker1)	0.12	0.33	-0.55	0.71
cor(Gesture2,Speaker1)	-0.09	0.32	-0.66	0.55
cor(Intercept,Gesture1:Speaker1)	-0.17	0.26	-0.64	0.36
cor(Gesture1,Gesture1:Speaker1)	0.03	0.32	-0.58	0.63
cor(Gesture2,Gesture1:Speaker1)	0.24	0.27	-0.32	0.72
cor(Speaker1,Gesture1:Speaker1)	-0.06	0.32	-0.65	0.56
cor(Intercept,Gesture2:Speaker1)	0.02	0.32	-0.60	0.63
cor(Gesture1,Gesture2:Speaker1)	0.03	0.33	-0.60	0.65
cor(Gesture2,Gesture2:Speaker1)	-0.05	0.33	-0.65	0.58
cor(Speaker1,Gesture2:Speaker1)	-0.03	0.33	-0.65	0.60
cor(Gesture1:Speaker1,Gesture2:Speaker1)	-0.03	0.33	-0.64	0.61

Table 27: Random-effects of the Speaker Bayesian beta binomial regression model with 95% credible intervals.

10.14 Model 3: experiment III

Coefficient	posterior mean	Standard Error	l-95% CrI	u-95% CrI
μ intercept	-1.60	0.11	-1.83	-1.38
μ distance (px)	0.50	0.07	0.36	0.65
ϕ intercept	2.06	0.14	1.78	2.33
ϕ distance (px)	-0.55	0.11	-0.76	-0.34

Table 28: Population-level estimates of the Bayesian beta binomial regression model on the log-odds scale with 95% credible intervals. The continuous predictor Distance (px) was standardised.

10.14.1 Posterior distributions over population-level estimates

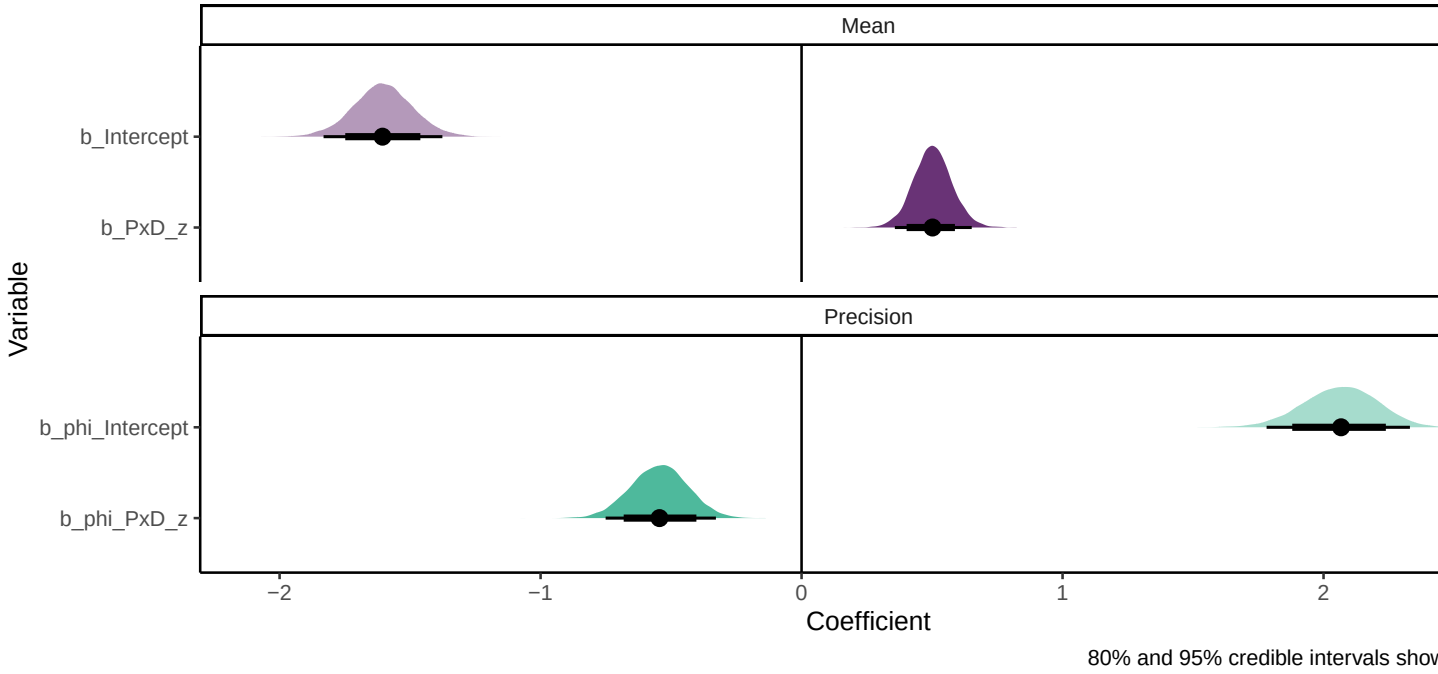


Figure 22: Posterior distribution over population-level estimates with 80% and 95% credible intervals.

10.15 Random effects estimates

Coefficient	posterior mean	Standard Error	l-95% CrI	u-95% CrI
Item (Number of levels: 20)				
sd(Intercept)	0.32	0.10	0.15	0.53
sd(PxD_z)	0.17	0.09	0.01	0.36
cor(Intercept,PxD_z)	-0.38	0.36	-0.90	0.45
Participant (Number of levels: 193)				
sd(Intercept)	0.88	0.09	0.71	1.05

Table 29: Random-effects of the Pixel Distance Bayesian beta binomial regression model with 95% credible intervals.